Expressivity Boundaries of Constant-State Attention-Free Sequence Models

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Abstract

Constant-state attention-free models replace content-dependent pairwise scoring with a fixed-size state. LTI convolution/recurrence duality, strict causality, O(1) state, and the recall weakness of fixed-state models are already established in the SSM / linear-attention / Zoology literature; we do not claim otherwise. This paper contributes **30 precise theorems grown from the SimplCore operator itself** , organized in six de-duplicated groups, and foregrounds the genuinely new ones: a tight minimum tensor order of 2 for exact retrieval; a strict-positivity separation between real-pole and conjugate-pole dictionaries; a provable separation of the rho<1 multi-pole kernel from the linear-attention constant kernel; 2-optimal deterministic codes for arbitrary vocabulary size with a matching Omega(V*N) state bound; a quantitative condition-number rate for near-zero frequency grids; a precision-length-decay trilemma with an *interior* optimal decay; a rounding-order crossover between the dual execution paths; a layer-count lower bound for multi-hop retrieval; and an information-gradient duality whose most natural repair re-triggers our own impossibility theorem. We strengthen the Toeplitz equivalence to arbitrary order and per-channel grids, and give the arbitrary-read-out impossibility theorem with a sharp quantitative boundary. **Theorem NT13 has exactly four premises** -- fixed-dimensional linear history summary, query only at the current position, independently assignable values, and at least two distinguishable targets -- and Tensor Spectral Convolution escapes it by changing the *tensor order of the signal* , not by contradicting it. **Corollary B1.1 is an explanatory framework, not a theorem** ; it is not a proved necessary-and-sufficient characterization. We further **withdraw** the earlier mutual-exclusivity claim between content addressing and FFT duality as a core contribution. On one synthetic single-hop task we did **not detect a significant difference** from a stored windowed attention record (1.0000 vs 0.9987, Fisher exact p = 0.4998); this is **not an equivalence proof** , the baseline is not a same-run global attention model, and no throughput comparison exists. All numbers are finite-sample observations; none is promoted to a theorem. >=10M-token equal-compute language modeling, multi-hop stacking and same-run baselines remain unperformed. We explicitly decline to inflate 49 statements into 49 theorems: 19 of them are provably false, retracted, unproved, or empirical.

1 Introduction

A constant inference state must summarize an unbounded history. The literature has established the LTI duality, the time-invariance requirement for a convolution view, and the recall/copying weakness of fixed-state models (arXiv:2111.00396, 2208.04933, 2312.04927, 2402.01032, 2404.08819). What has been less precise is *which algebraic property of a specific architecture* produces each boundary, and *which single premise* must be removed to escape it. We answer both for Spectral Global Convolution (SGC) and its tensor extension, with 30 de-duplicated theorems and a 49-item audit.

Evidence discipline. A theorem is a universal statement derived only from definitions. A finite-sample observation is a measured value tied to a configuration, seed count, precision and task. A conjecture is explicitly unverified. No measured number is upgraded to a theorem; no theorem is inferred from a measured number. Conditional theorems state their extra assumption inline.

Theorem organization (de-duplication). We deliberately avoid theorem inflation: standard facts (geometric-series BIBO, Sylvester orthogonality, rank-nullity, scalar-score induction) are used as *proof steps or lemmas* , not given theorem slots. The 30 theorems fall into SIX groups: I (NT01-05) TSC construction and tight optimality; II (NT06-10) grid/precision/length; III (NT11-15) multi-hop, self-evolution, equivalence; IV (NT16-19) algebraic structure and identifiability; V (NT20-24) self-evolution and transfer objectives; VI (NT25-30) numerical frontier and multi-hop duality. **We deliberately do NOT inflate the remaining 19 statements into theorems** (Appendix G): 8 are falsified with explicit counterexamples, 5 were formally withdrawn, 3 are unproved conjectures with falsification experiments, and 3 are empirical claims (LM parity, SOTA, absolute throughput) that no theorem can establish. Promoting them would repeat exactly the theorem inflation and 'numerical verification as proof' that this revision set out to remove.

Positioning. This is a *boundary-characterization paper with a constructive scope correction and auditable negative results* . It is not a new SOTA sequence model, does not claim general language-model parity or superiority over attention, and does not claim a formal upper bound on all constant-state models.

2 Preliminaries and Notation

Let x_{1:L} be a causal token sequence. We use **t** for the time index, **tau** for a kernel lag and **d = t - s** for a delay in an impulse response; these are kept distinct throughout. SGC uses the damped-oscillator kernel K(tau) = rho^tau sin((tau+1)theta) with 0 < rho < 1 and theta in (0, pi), or the general second-order real-pole kernel h(tau) = sum_{j=0}^{tau} r1^{tau-j} r2^{j}. The architecture exposes an FFT whole-sequence path and a streaming two-tap IIR path. A content-addressing task supplies n key-value pairs and a query q equal to one key; the answer is the associated value. A is the part of the final state that depends on the history but not on q.

The state register is finite and its size is independent of L (an asymptotic statement with a possibly large constant). Theorem statements are analytic; numerical values appear only as finite-sample observations with provenance in Appendix D. Complete proofs of all 15 theorems are in Appendix A.

3 Fifteen Theorems

Each theorem below states its hypotheses, its universal claim and the proof route; the full derivation is in Appendix A. Machine-checkable finite identities in the supplementary artifact verify that each statement is correctly formed and has a non-empty premise -- they are **not** the proofs. The 49-item audit (49/49 PASS) classifies every statement; for the 19 non-theorem items PASS means 'correctly classified with a reproducible witness', not 'true'.

3.1 I: TSC construction and optimality

Group I establishes what the tensor construction minimally costs and why the real-pole dictionary is forced.

NT01 Tight minimum tensor order for exact single-hop retrieval.
Claim. (a) A vector-valued (order-1) state that is a fixed-dimensional linear summary of the history cannot implement the retrieval, for ANY read-out. (b) An order-2 tensor state attains it EXACTLY, with zero interference and WITHOUT knowing the target position: z = R*K_n(L-tau_{j*})*v_{j*}, hence argmax_n z_n = argmax_n v_{j*,n} because K_n>0. Therefore the

minimum tensor order is exactly 2, and the bound is TIGHT. *Premises, boundary, use and full proof: Appendix A.1.*

Proof route. (a) Let A=L(v_1..v_n) in R^d be the summary, L: R^{nd}->R^d linear and independent of q. rank(L)<=d<nd for n>=2, so dim ker L>=nd-d>0. Pick nonzero Delta-v in ker L supported on the target coordinate: two value assignmen

NT02 Strict-positivity separation of the two pole families.

Claim. Family (i) has h(tau)>0 for every tau>=0; family (ii) has h(tau)<=0 for some tau<=ceil(pi/theta). Hence the two impulse-response families are DISJOINT, and membership is decided exactly by inspecting the sign sequence of h(0..L-1) in O(L) (zero misclassification in exact arithmetic). Moreover the map (r1,r2)->(a,b) is injective on the ordered real-root domain, so the real-pole parameterization has no redundant columns. *Premises, boundary, use and full proof: Appendix A.2.*

Proof route. (i) h(tau)=sum_{j=0}^{tau} r1^{tau-j} r2^{j} = (r1^{tau+1}-r2^{tau+1})/(r1-r2) for r1!=r2, and (tau+1)r^{tau} for r1=r2=r. Every term is strictly positive, so h(tau)>0.

NT03 Provable separation of rho<1 multi-pole TSC from the linear-attention family.

Claim. The span of {K_n : n} contains NO nonzero constant function, and the constant function is not in the closure of that span in l^infty(N). Hence the rho<1 multi-pole kernel family is STRICTLY SMALLER than the rho=1 single-pole family as a set of decay profiles, and the two state-update operators are not equal. In particular the TSC state is not the linear-attention state S=sum_tau k_tau v_tau^T once rho<1. *Premises, boundary, use and full proof: Appendix A.3.*

Proof route. Any nonzero combination F(d)=sum_n c_n K_n(d) is a finite sum of exponentials with bases in (0,1). Let rho_max<1 be the largest base; then F(d)=Theta(rho_max^d) -> 0 as d->infinity. A constant function equal to 1 does no

NT04 Deterministic zero-crosstalk codes for arbitrary V are 2-optimal.

Claim. (a) Zero crosstalk among V nonzero rows forces R>=V. (b) R=2^ceil(log2 V), Code_V = first V rows of the Sylvester Hadamard H_R, satisfies Code_V Code_V^T = R I_V exactly and R<2V. Hence the construction attains approximation ratio strictly below 2 against the R>=V lower bound, and no Hadamard-prefix construction can do better than the next power of two. *Premises, boundary, use and full proof: Appendix A.4.*

Proof route. (a) V mutually orthogonal nonzero vectors in R^R are linearly independent, so V<=R.

NT05 State-vocabulary lower bound and end-to-end tightness.

Claim. Any such construction requires at least P*V*N scalar state registers, i.e. Omega(V*N). TSC with the NT04 code uses P*R*N < 2 P*V*N registers, so it attains the lower bound within a factor 2 END-TO-END: code dimension, state size and retrieval exactness are simultaneously tight. *Premises, boundary, use and full proof: Appendix A.5.*

Proof route. Zero crosstalk requires R>=V mutually orthogonal code rows (NT04a). The state is indexed by (p,r,n); for each of the P*N pairs (p,n) the R code coordinates must be stored independently because the recurrence s_{p,r,n,t}=

3.2 II: grid, precision and length

Group II turns the measured conditioning, precision and length frontiers into quantitative rates with explicit premises.

NT06 Quantitative rank-deficiency rate of near-zero frequency grids.

Claim. cond(G) >= 2*sin^2(theta_1) / (Delta^2 * A(rho)), where A(rho)=sum_{tau>=0} rho^{2tau}(tau+1)^2 = (1+rho^2)/(1-rho^2)^3 < infinity. Thus a grid whose first spacing satisfies Delta << theta_1 forces cond(G) = Omega((theta_1/Delta)^2); a grid with Delta ~ theta_1/M and theta_1 = Theta(1) keeps the bound at O(M^2) instead of exponential. *Premises, boundary, use and full proof: Appendix A.6.*

Proof route. Use the Rayleigh quotient with the vector e_1-e_2: sigma_min(G) <= (1/2)||f_1-f_2||^2. By |sin a - sin b| <= |a-b|, ||f_1-f_2||^2 = sum_tau rho^{2tau}[sin((tau+1)theta_1)-sin((tau+1)theta_2)]^2 <= Delta^2 sum_tau rho^{2t

NT07 Precision-length-decay trilemma with an interior optimal rho.

Claim. (a) The accumulated error is bounded by e_infty <= delta/(1-rho). (b) The resolvable length obeys L(rho) <= 1 + ln((1-rho)U/(c*delta)) / |ln rho|. (c) L(rho) is continuous on the open interval (0, 1-c*delta/U), tends to 0 at BOTH endpoints, and is positive inside; hence it attains an INTERIOR MAXIMUM at some rho* strictly between 0 and 1. Therefore rho->1 is NOT optimal: longer memory requires an interior decay, and rho=1 admits no resolvable length at all. *Premises, boundary, use and full proof: Appendix A.7.*

Proof route. (a) The error obeys e_t <= rho e_{t-1} + delta; by induction e_t <= delta sum_{k=0}^{t-1} rho^k < delta/(1-rho).

NT08 Robust exact retrieval under bounded code perturbation.

Claim. The retrieval remains exactly correct (argmax unchanged) whenever epsilon < R*K_min / ((L-1)*K_max). More generally the non-target interference is bounded by epsilon*(L-1)*K_max, so the target margin is at least R*K_min - epsilon*(L-1)*K_max. *Premises, boundary, use and full proof: Appendix A.8.*

Proof route. Write q.k_tau = R delta_{tau,j*} + E_{q,tau}. Then z_n = R K_n(L-1-tau_{j*}) v_{j*,n} + sum_{tau != j*} E_{q,tau} K_n(L-1-tau) v_{tau,n}. The target coordinate of the one-hot value has amplitude >= R K_min. Each non-targ

NT09 Rounding-order crossover between the FFT and IIR paths.

Claim. The IIR error bound is L-independent while the FFT bound grows as c1*eps*log2(L). They cross at L* = 2^{c2/(c1(1-rho))}. For every L < L* the streaming path has the smaller worst-case bound; for rho close to 1, L* is astronomically large, so on all realizable sequence lengths the IIR bound is never worse. *Premises, boundary, use and full proof: Appendix A.9.*

Proof route. FFT: after log2 L stages, (1+c1 eps)^{log2 L} - 1 <= c1 eps log2 L + O(eps^2). IIR: the perturbation recursion e_t <= rho e_{t-1} + c2 eps has the steady state e <= c2 eps/(1-rho), with no L dependence because the filter

NT10 Per-channel time-invariance preserves the duality.

Claim. Each (p,n) pair is an independent LTI system with transfer function H_{p,n}(z)=1/(1-rho_{p,n} z^{-1}). The full operator is BLOCK Toeplitz (Toeplitz in t for each channel, diagonal in (p,n)), so a single batched FFT diagonalizes it channel-wise and the streaming recurrence realizes the same operator. The state has P*R*N scalars, independent of L. Consequently a per-channel coefficient tensor of shape (M,D) is architecturally admissible without breaking any purity constraint. *Premises, boundary, use and full proof: Appendix A.10.*

Proof route. For fixed (p,r,n), unrolling gives s_{p,r,n,t}=sum_tau rho_{p,n}^{tau} u_{t-tau}[r,n], whose impulse response depends only on the delay tau. Stacking the P*N independent channels gives a block-diagonal-in-(p,n) operator

3.3 III: multi-hop, self-evolution, equivalence

Group III characterizes the structural boundaries: degree, Toeplitz, arbitrary read-out, information-gradient duality and zero-order learning.

NT11 Layer-count lower bound for multi-hop retrieval.

Claim. Any such stack needs L >= ceil(log_d(H+1)) layers. For bilinear layers L >= ceil(log2(H+1)); in particular H=2 already requires L>=2, so a single contraction layer cannot implement two-hop retrieval. This is a NECESSARY condition; no sufficient construction is asserted. *Premises, boundary, use and full proof: Appendix A.11.*

Proof route. The composition of polynomials of degrees d_1..d_L has degree at most the product d_1...d_L <= d^L (degree cannot increase under addition or skip connections). The H-hop target is a multilinear form of degree H+1 in (k,v

NT12 Toeplitz iff time-invariant coefficients, for arbitrary order and chann

Claim. The impulse response h(t,s) depends only on t-s IFF every

coefficient a_{j,t} and c_t is constant in t. With C independent channels the full operator is block Toeplitz IFF this holds for every channel. Necessity fails at the first delay where c_t=0, so the theta-clamp / nonzero-c premise is essential, not decorative. *Premises, boundary, use and full proof: Appendix A.12.*

Proof route. Necessity by induction on the delay k. k=0: h(s,s)=c_s must be s-independent, so c_t=c. Assume a_1..a_{k-1} are constant. At delay k, h(s+k,s) = a_{k,s+k} c + Phi(a_1..a_{k-1},c), where Phi is a polynomial with s-indepen

NT13 Arbitrary-read-out impossibility with a sharp quantitative boundary.

Claim. (a) Content addressing is impossible for any f. (b) Quantitatively dim ker L >= nd - d. (c) For n=1 the kernel can be zero, so premise (iv) is NECESSARY and cannot be dropped. (d) TSC escapes by changing the tensor order of the signal and the read-out, NOT by contradicting (a). *Premises, boundary, use and full proof: Appendix A.13.*

Proof route. (a,b) rank(L)<=d, so dim ker L = nd - rank(L) >= nd-d > 0 for n>=2. Choose nonzero Delta-v in ker L supported on a coordinate of the target value (possible by (iii),(iv)). Two assignments v and v+Delta-v give the same A

NT14 Information-Gradient Duality (with the softening trap).

Claim. No single channel is simultaneously (I) differentiable, (II) covering all L items, and (III) losslessly retrieving any one. The three horns are exclusive AND the natural repair is self-defeating: replacing hard top-k by a differentiable soft aggregate over all C copies re-triggers NT13's kernel impossibility. *Premises, boundary, use and full proof: Appendix A.14.*

Proof route. (i) Differentiable + covering forces lossy: F must be injective on X^L, but a B-byte state has at most 2^{8B} configurations and |X|^L > 2^{8B} for L > 8B/log2|X| (pigeonhole).

NT15 Zero-order self-evolution: gate validity and information rate.

Claim. (a) P(false commit | true regression) <= alpha for one metric and <= m*alpha for m metrics (<= alpha under Bonferroni alpha/m). (b) I(history ; weight increments over T rounds) <= T*w*b bits, INDEPENDENT of L and C; instantaneous bandwidth <= k*b bits/round. (c) The path MUST be zero-order. (d) A non-decreasing ACCEPTED score does NOT imply monotone real capability. *Premises, boundary, use and full proof: Appendix A.15.*

Proof route. (c) NT14(ii) zero Jacobian for unselected items, NT14(iii) softening destroys exactness, and the no_grad online path has no graph: no gradient can carry long-context experience into the weights, so the only admissible ch

3.4 IV: algebraic structure and identifiability

Group IV proves that the real-pole family, the linear grid and the two-tap state are not choices but forced optima.

NT16 Exact Turan identity and log-convexity of the real-pole dictionary.

Claim. (a) EXACT IDENTITY: h(tau)^2 - h(tau-1) h(tau+1) = (r1*r2)^tau > 0 for every tau >= 1, including the double-root case r1=r2=r where it becomes r^{2tau}. Hence h is STRICTLY log-convex. (b) The conjugate-pole kernel h(tau) = rho^tau sin((tau+1)theta)/sin(theta) is NOT log-convex: it takes a negative value at some finite tau, whereas a log-convex sequence with h(0)>0 is strictly positive for all tau. (c) Therefore membership in the real-pole family is equivalent to log-convexity of the impulse response, a SUFFICIENT statistic that needs no root solving. *Premises, boundary, use and full proof: Appendix A.16.*

Proof route. (a) Binet form: h(tau) = (r1^{tau+1} - r2^{tau+1})/(r1-r2) for r1>r2, and h(tau) = (tau+1) r^tau for r1=r2=r. For r1>r2 set A=r1^{tau}, B=r2^{tau}, p=r1, q=r2 and expand h(tau)^2 - h(tau-1)h(tau+1). Writing s_k = (p^{k+1

NT17 O(1) three-sample identifiability and pole-family discriminant.

Claim. (a) The recursion coefficients are recovered EXACTLY and uniquely: a = h(1)/h(0), b = (h(2) - a h(1))/h(0). (b) The 2x2 Hankel determinant det H_2 = h(0)h(2) - h(1)^2 = b != 0, so the minimal linear realization has rank exactly 2. (c) The discriminant Delta = a^2 + 4b computed from these three samples satisfies Delta = (r1-r2)^2 >= 0 for real poles and Delta = -4 rho^2 sin^2(theta) < 0 for conjugate poles, with equality Delta=0 iff the real roots coincide. Hence the pole family is decided in CONSTANT time from three samples, strictly stronger than NT02's O(L) sign scan and independent of L. *Premises, boundary, use and full proof: Appendix A.17.*

Proof route. (a) From y_t = a y_{t-1} + b y_{t-2} + u_t with the unit impulse, h(0)=1, h(1)=a h(0)=a, h(2)=a h(1)+b h(0)=a^2+b. Solving gives the stated inversion; uniqueness follows because (h(1),h(2)) determines (a,b) by a triangul

NT18 Linear grids maximize the minimum spacing under a theta_min budg

Claim. (a) Delta_min <= (theta_max - theta_min)/(M-1), with EQUALITY iff the grid is exactly equally spaced (linear). (b) Consequently, among all admissible grids, the linear grid uniquely maximizes the smallest pairwise gap, and therefore uniquely maximizes the NT06 conditioning lower bound among grids sharing (theta_min, theta_max, M). (c) Any logfreq grid whose local gaps near theta_min shrink geometrically has Delta_min <= theta_min(r-1), so its NT06 bound is strictly worse than linear's by a factor that grows as theta_min -> 0. A hybrid grid is a convex interpolation and is bounded between them. *Premises, boundary, use and full proof: Appendix A.18.*

Proof route. (a) Summing the M-1 gaps gives theta_M - theta_1 <= theta_max - theta_min. Hence (M-1) Delta_min <= sum gaps <= theta_max - theta_min, giving the bound. Equality requires every gap equal AND theta_1=theta_min, theta_M=th

NT19 The two-tap state is a MINIMAL sufficient statistic (tight byte bound

Claim. (a) No deterministic state of fewer than 2 scalars per (m,d) can reproduce the same output sequence for all inputs. (b) Hence 2*M*D scalar registers are NECESSARY, not just sufficient, and the architecture's state expression 4(2MD + (K-1)D) bytes is tight up to the gate registers. (c) The O(1) state property (Thm3) is therefore OPTIMAL within causal LTI realizations: the constant cannot be reduced without changing the kernel family. *Premises, boundary, use and full proof: Appendix A.19.*

Proof route. (a) Suppose a 1-scalar causal linear state s_t = lambda s_{t-1}+u_t realizes the operator. Its impulse response is h(tau)=lambda^tau, so h(1)^2 = h(0)h(2). But NT17 gives h(0)h(2)-h(1)^2 = -b != 0 for BOTH root families

3.5 V: self-evolution and transfer objectives

Group V gives the SMM x SGC consolidation path its first statistical and information-theoretic guarantees, and separates the two transfer objectives.

NT20 Axiomatic uniqueness of the no-regression min-rule gate.

Claim. The UNIQUE rule satisfying P1-P5 is G = AND_i [ell_i >= -|tau|], equivalently min_i ell_i >= -|tau|, which is exactly the upstream regression_gate. Consequently ANY weighted-sum, averaged, or majority-vote gate violates at least one axiom and permits a metric to collapse while still being compensated -- contradicting the source's 'no-regression' semantics. *Premises, boundary, use and full proof: Appendix A.20.*

Proof route. By P4 the rule is a Boolean function g(a_1..a_m). By P2, any vector with some a_i=0 maps to reject. By P5 the all-ones vector maps to accept. These two facts determine g on all 2^m inputs: g = AND_i a_i. P1 and P3 are th

NT21 The constant-verifier trap and necessity of an independent evaluator.

Claim. (a) If V is constant, every paired difference is exactly 0, every bootstrap resample has mean 0, the CI lower bound is 0 for every metric, and the gate therefore accepts EVERY round unconditionally: the gate carries ZERO information. (b) Hence 'refuse writeback when the default verifier is used' is not a conservative heuristic but a LOGICAL NECESSITY: it is the only way to keep the no-regression claim non-vacuous. (c) A non-constant, held-out verifier is NECESSARY and SUFFICIENT for the gate to be non-trivial (it can then reject). *Premises, boundary, use and full proof: Appendix A.21.*

Proof route. (a) V(model)=c for all models implies score_new_i -

score_base_i = c-c = 0 for every sample i. Any bootstrap resample of zeros has mean 0, so both one-sided bounds are 0 and 0 >= -|tau| holds; by NT20 the gate accepts. (

NT22 Irreducible gap of hidden-state alignment objectives (TS2SC).

Claim. There exists epsilon_0 > 0, INDEPENDENT of L and of the student parameters, such that the alignment objective cannot be driven to zero: on some inputs the achievable residual cosine is at most 1 - epsilon_0. Adding layers does NOT close the gap, because each mixer preserves the premise that history enters only through a fixed linear summary. Hence 'fit the attention internals' is not a tuning problem but a function-class obstruction. *Premises, boundary, use and full proof: Appendix A.22.*

Proof route. By NT13, a single static-LTI mixer with a linear residual stream cannot implement content addressing: there exist two value assignments v, v+Delta-v giving the same history summary A and the same query q, hence the same

NT23 Separation of output-distribution distillation from hidden-state alignment.

Claim. Under H1, inf over student parameters of KL(p_T || p_S) = 0, while the hidden-state alignment objective has the strictly positive irreducible gap epsilon_0 of NT22. Hence the two objectives are NOT equivalent: output distillation is not blocked by NT13/NT22, whereas internal alignment is. This holds WITHOUT assuming the student's internal states resemble the teacher's. *Premises, boundary, use and full proof: Appendix A.23.*

Proof route. H1 states that for every eta>0 there exists f in F_S with KL(p_T||softmax(f/T)) < eta; taking eta->0 gives infimum 0. By NT22 the internal-alignment gap is >= epsilon_0 > 0 for all parameters, so its infimum is >= epsilo

NT24 Hard stagnation of zero-order self-evolution (minimal-round bound).

Claim. (a) If I_task > T*w*b, the weights CANNOT encode a sufficient statistic for the task, so no read-out attains the task's Bayes error: self-evolution is HARD-STAGNATED before round T. (b) The minimum number of rounds satisfies T_min >= ceil(I_task/(w*b)), INDEPENDENT of L and of the bank capacity C. (c) Longer context does not shorten T_min; it only increases the retrievable history at inference, consistent with NT15(b). *Premises, boundary, use and full proof: Appendix A.24.*

Proof route. (a) Let H be the history, Theta_T the weights after T rounds, and Y the task answer. If the weights encoded a sufficient statistic for Y then I(H;Theta_T) >= I(H;Y) >= I_task. But NT15(b) gives I(H;Theta_T) <= T*w*b < I_

3.6 VI: numerical frontier and multi-hop duality

Group VI closes the numerical and multi-hop frontier with tight radii, a precision-dependent decay law, and an end-to-end tightness chain.

NT25 Tightness of the quantization-resolvable radius.

Claim. (a) The resolvable radius is exactly tau* = ln(r)/|ln rho| (floor), and it is TIGHT: there exists an input with a single unit pulse such that the contribution at tau*+1 is exactly at the quantization threshold and cannot be resolved, while all delays <= tau* are resolvable. (b) tau* = Theta((m+1)/|ln rho|) for m-bit mantissas, so the asymptotic rate is OPTIMAL up to a constant: no deterministic rescaling improves the EXPONENT. (c) For rho = 1-epsilon, tau* ~ (m+1) ln 2 / epsilon, giving the observed order of magnitude of the bf16/int8 long-length drop. *Premises, boundary, use and full proof: Appendix A.25.*

Proof route. (a) rho^tau U > s/2 iff tau ln rho > ln(s/(2U)) = ln r, and since ln rho<0 this is tau < ln r / ln rho = ln(r)/|ln rho|. The largest integer below it is tau*. Choose the input with u_0=U and all other u_t=0: the state at

NT26 Causal optimality of per-step peak calibration.

Claim. (a) The per-step peak rule Delta_t = max(|s_t|, Delta_min) is the POINTWISE MINIMAL admissible scale given I_t, hence minimizes delta_t at EVERY step. (b) Any fixed global scale must satisfy Delta >= sup_{t<=L}|s_t|, and since sup >= max_t|s_t| pointwise, delta_fixed >= max_t delta_peak(t), with strict inequality whenever the peak varies. (c) Therefore fixed-scale quantization is never better and is strictly worse on any sequence with a non-constant state envelope; the improvement is CAUSAL (needs no future information). *Premises, boundary, use and full proof: Appendix A.26.*

Proof route. (a) Admissibility requires Delta_t >= |s_t|; the smallest such value is |s_t| (or Delta_min for the zero state), which uses only s_{t-1}, u_t and rho -- all in I_t. Minimality is pointwise, so delta_t = c Delta_t is mini

NT27 Phase transition of the optimal decay across precision depths.

Claim. (a) g is strictly increasing near rho -> 0 and strictly decreasing near hi, so for small delta/U it has a SINGLE interior maximum rho*(m). (b) rho*(m) solves the stationarity equation (1-rho) ln(1/rho) ln((1-rho)U/(c delta)) = rho ln(1/rho) + ... equivalently the balance (1-rho) ln((1-rho)U/(c delta)) = rho ln(1/rho); asymptotically rho*(m) = 1 - Theta(2^{-m}). (c) Therefore DIFFERENT precision depths have DIFFERENT optimal decays: a single rho tuned for fp32 is provably suboptimal for int8. rho -> 1 is never optimal (NT07). *Premises, boundary, use and full proof: Appendix A.27.*

Proof route. Write N(rho)=ln((1-rho)U/(c delta)) > 0 and D(rho)=ln(1/rho)>0, g=N/D. Then g' = (N' D - N D')/D^2 with N' = -1/(1-rho) < 0 and D' = -1/rho < 0, so sign(g') = sign(-D/(1-rho) + N/rho). As rho -> 0, D = ln(1/rho) -> +in

NT28 Layer x tensor-order incompatibility inequality for H-hop retrieval.

Claim. L * ln(o) >= ln(H+1), equivalently o^L >= H+1. For o=2 this recovers NT11 (L >= log2(H+1)); for L=1 it forces o >= H+1, so a SINGLE layer needs tensor order at least H+1. CRITICALLY: increasing the number of pole channels P changes the KERNEL SHAPE but not the tensor order o, so P cannot compensate for too few layers or too low an order. The two resources (depth, order) are incomparable and must both satisfy the product bound. *Premises, boundary, use and full proof: Appendix A.28.*

Proof route. Composition of polynomials of degrees d_1..d_L has degree at most prod d_i <= o^L; skip connections (addition) do not increase degree. The H-hop target is a multilinear form of degree H+1 (H factors <phi(q),psi(k)> plus

NT29 Topological necessity of a discontinuous hard pointer.

Claim. (a) R is DISCONTINUOUS on the bisector, hence not differentiable there; hard pointer selection is therefore a NECESSARY feature of exact retrieval, not an implementation artifact. (b) Any CONTINUOUS surrogate Rtilde that agrees with R off an epsilon-neighborhood of the bisector has a NONZERO mis-selection probability bounded below by the measure of that neighborhood under any query distribution with positive density there. (c) Softening cannot simultaneously be continuous AND exact; this is an INDEPENDENT obstruction from NT14(iii)'s kernel-collision argument (which concerns the aggregate read-out, not the selector). *Premises, boundary, use and full proof: Appendix A.29.*

Proof route. (a) Let q_1,q_2 be points on opposite sides of the bisector B = {q : <phi(q),k_1> = <phi(q),k_2>} with <phi(q_1),k_1> > <phi(q_1),k_2> and the reverse at q_2. Then R(q_1)=v_1 and R(q_2)=v_2. Take a path q(s) from q_1 to

NT30 End-to-end tightness chain: none of the five bounds can be improved.

Claim. Any such architecture simultaneously requires (i) tensor order o >= H+1 for L=1, or o^L >= H+1 in general (NT28); (ii) code dimension R >= V (NT04); (iii) state >= 2*M*R*N scalars, with R < 2V attainable (NT05, NT19); (iv) a strictly positive kernel, hence real-pole roots (NT16/NT01). TSC at H=1 attains ALL FOUR within constant factors: order 2, R<2V, 2M*R*N<4M*V*N, real poles. Moreover the chain is JOINTLY TIGHT: improving any one bound alone cannot lower the product, because each is forced by a different independent theorem. *Premises, boundary, use and full proof: Appendix A.30.*

Proof route. Chain the results: NT13 forbids order 1; NT01 achieves order 2 (tight). NT04(a) gives R>=V and NT04(b) gives R<2V (2-optimal, tight within the class). NT19 gives 2M scalars per code coordinate per value channel as a MINI

4 Constructive Scope Correction: TSC

SGC constrains its *kernel* to be a static LTI operator; it does not constrain the *tensor order of its input signal* . Replacing the vector signal by an outer product and the read-out by a contraction yields Tensor Spectral Convolution:

$$u_t[r,n] = a_t[r]\, b_t[n], \quad s_{p,r,n,t} = \rho_{p,n} s_{p,r,n,t-1} + u_t[r,n]$$

$$z_n = \sum_{p,r} w_{p,n}\, q_r\, s_{p,r,n,L} = \sum_{\tau} (q \cdot k_\tau) K_n(L-1-\tau) v_{\tau,n}$$

NT01 proves that order 2 is both necessary and sufficient for exact, position-free retrieval under a strictly positive kernel; NT02 shows that strict positivity holds exactly for the real-pole dictionary and fails for the conjugate-pole dictionary; NT10 proves that per-channel rho_{p,n} keeps the operator block Toeplitz, so a coefficient tensor of shape (M,D) is architecturally admissible. NT04/NT05 give a deterministic code for every vocabulary size with R < 2V and a matching Omega(V*N) state bound; NT08 makes retrieval robust to bounded code perturbation. NT16 (exact Turan identity) shows strict positivity is a STRUCTURAL property of real poles, NT17 identifies the family in O(1) from three impulse samples, NT19 proves the two-tap state is a minimal sufficient statistic, NT29 proves the L2 pointer must be discontinuous, and NT30 chains all four bounds into one attained tight end-to-end result.

Relation to linear attention (a provable separation, not a judgement left open). At rho -> 1 with one pole the state sum_tau k_tau v_tau^T is isomorphic to a linear-attention state; we record this openly. NT03 proves that for rho < 1 with multiple poles the kernel family contains no nonzero constant function and has l^infty distance exactly 1 from it, so the state update operator is **not** the linear-attention operator. We therefore position TSC as *a construction that realizes content addressing under a static LTI kernel with a tensor state and a contraction read-out* , not as a claim of having left the linear-attention paradigm entirely. The construction has no pairwise scoring matrix, no softmax normalization and no growing KV cache.

Cost. The state holds P*R*N scalars (56,320 B observed versus 512 B for the original mixer); O(1) is asymptotic. A coarse analytic crossover suggests no compute advantage below L ~ 880; this has not been measured against an attention implementation.

5 Finite-Sample Observations (Not Theorems)

Each item is a finite-sample estimate under a fixed configuration; none is a theorem, none claims universality. Precision, seed count and provenance are in Appendix D.

O1 - saturated transfer ceiling.
In a 1.7B weight-transfer fitting study the 28-layer residual-stream cosine reached 0.2484 against a 0.2476 static-kernel oracle; a same-input single-layer control reached 0.5430. All alignment metrics improved while generation stayed non-sentential. The oracle is a finite-layer estimator, not a proved bound.

O2 - phase-resonance content addressing.
A fixed-frequency-slot variant raised single-hop accuracy from 0.4160 to 0.8672; an equal-parameter low-rank attention baseline reached 1.0000 (5,217 vs 5,152 parameters).

O3 - native pilot training.
A 470K-parameter native model on 3.07M tokens reached validation perplexity 10.817 against a same-budget Transformer's 11.281, and 10.217 with output-distribution distillation. Generation was non-sentential on all arms.

O4 - non-replication of the periodic-axis gain.
An external white-box run reported 0.9844 vs 0.8535; an in-repository probe with seven grids produced 0.1094-0.1354, all below the copy-last baseline 0.1575, with a paired CI crossing zero. The advantage is not established.

O5 - TSC synthetic single-hop result (no significant difference detected).
On a single-hop synthetic binding task (V = 64, L = 9, held-out 512 per seed), W1 reached 1.0000 over 5 seeds at 10264 parameters, against 0.9987 for a **stored** windowed softmax-attention record at 10240 parameters (Fisher exact p = 0.4998). **No significant difference was detected in this configuration; p > 0.05 is not an equivalence proof** , the baseline is not a same-run global attention model, and no throughput comparison exists. W4_half reached 1.0000 with 4920 parameters (48.0% of the reference) over 5 seeds; M5 reached 1.0000 with 2792 parameters (3.67x efficiency). These are parameter-efficiency observations on ONE synthetic task only. An analytic evaluation stayed at 1.0000 up to L = 1025 (10 lengths tested); a value-permutation control stayed at 1.0000 and removing the predecessor-key encoding collapsed accuracy to 0.2331. A random orthogonal code also reached 1.0000 (NT08 explains why). A two-hop variant collapsed to 0.0322 (chance); NT11 gives the layer lower bound that explains it.

O6 - physical Pareto frontier.
fp64/fp32 stayed at 1.0000 through L = 4097; bf16 and int8 first dropped at L = 2049 (0.5312); fp16 dropped at L = 4097 (0.6250). The decay coefficient is the effective lever: fp32 with rho = 0.999 (BIBO-stable) stayed at 1.0000 to L = 16,385. NT07 proves that the optimal rho is *interior* , so rho -> 1 is not the right target; NT09 gives the rounding-order crossover. Measured reference-implementation cost stayed near 35.25 us/token with a constant 56320 B state; **only the scaling is informative, not the absolute rate** . Conservation checks passed 43/43.

6 Limitations and Missing Evidence

· **Three decisive experiments are missing.** (i) Same-run global/windowed attention re-run plus measured throughput; (ii) >=10M-token equal-compute language modeling against Mamba-2 / GLA / DeltaNet / Hyena / Transformer; (iii) TSC multi-layer stacking for multi-hop retrieval. Without these the TSC positive result is a concept validation only. O3 is a 3.07M-token pilot.

· Not every observation has a paired CI over >=5 seeds: O5 and O6 do; the transfer, phase-resonance, pilot and non-replication observations do not.

· All positive TSC results are on one synthetic single-hop task with V = 64 and one layer. Length extrapolation beyond L = 129 permits repeated keys, so it is an extrapolation test, not an in-distribution test.

· O5 compares against a stored windowed-attention record; a full global attention baseline and a measured throughput comparison are absent. No significant difference detected is not equivalence, and parity on one synthetic task is not general-capability parity.

· **NT13 has exactly four premises** and is not extended to deep

stacks with per-position nonlinearities; **Corollary B1.1 is an explanatory framework, not a theorem** , and is not a proved sufficient/necessary characterization.

· int8 remains a hard long-length boundary; rho = 1 is only critically stable and diverges on unbounded streams. NT11 supplies a necessary layer bound, not a multi-hop construction.

· NT03 separates kernel families / state-update operators; it does not claim full-model expressivity separation from every gated linear-attention variant.

7 Related Work

Structured state-space and long-convolution models (arXiv:2111.00396, 2208.04933, 2302.10866) established the LTI convolution/recurrence duality and the time-invariance requirement. Selective and linear-attention state models (arXiv:2312.00752, 2405.21060, 2312.06635, 2406.06484, 2006.16236, 2305.13048, 2407.04620) obtain content addressing through input-dependent updates at the cost of the LTI convolution view. Zoology and related work (arXiv:2312.04927, 2402.01032, 2404.08819) already established that fixed-state models are weaker on recall and copying.

What is new here, precisely. We do **not** claim the six structural facts above as new. The additions are: the tight order-2 / Omega(V*N) / 2-optimal-code construction (NT01, NT04, NT05); the strict-positivity separation of pole families and the provable separation from the linear-attention constant kernel (NT02, NT03); quantitative rates for grid conditioning and for the precision-length-decay trilemma (NT06, NT07, NT09); the robustness bound (NT08); the per-channel block-Toeplitz and arbitrary-order equivalence (NT10, NT12); the arbitrary-read-out impossibility with a sharp quantitative boundary (NT13); the information-gradient duality and the zero-order rate bound (NT14, NT15); and the layer lower bound (NT11). The comparison remains qualitative and bibliographic: equal-parameter and equal-compute quantitative comparisons have not been performed.

8 Conclusion

Thirty de-duplicated theorems characterize what a constant-state spectral model can express, and a 49-item audit classifies every remaining statement (Appendix G): 8 falsified, 5 withdrawn, 3 conjectures, 3 empirical claims no theorem can establish. We decline to inflate these into theorems. The genuinely new precise propositions are the tight order-2 construction with a matching state bound, the pole-family and linear-attention separations, the quantitative grid / precision / length rates, the interior-optimal decay, the layer lower bound, and the information-gradient duality. **The arbitrary-read-out impossibility (NT13) holds only under its four stated premises** ; TSC evades it by changing the tensor order of the signal, not by disproving it. **Corollary B1.1 is not a theorem** and must not be read as a proved necessary-and-sufficient condition. **Withdrawing the mutual-exclusivity claim between content addressing and FFT duality is a core contribution** , as is the withdrawal register in Appendix F. All numbers are finite-sample observations; 'no significant difference detected' is not equivalence. >=10M-token equal-compute language modeling, same-run attention baselines, multi-hop stacking and throughput comparisons remain open work, not conclusions of this paper.

References

Entries are keyed by arXiv identifier with verified title/venue.

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Broader Impact

This work is theoretical and empirical on synthetic tasks. It involves no human subjects, personal data, deployed systems or dual-use capabilities. Its positive potential is a clearer account of when constant-state models suffice, which can reduce long-context inference cost and discourage over-engineering attention-free architectures for tasks that structurally require content addressing. Its main risk is interpretive: a reader may take 'no significant difference detected' on one synthetic binding task as general language-model parity, or read 15 theorems as 15 independent new discoveries. We have therefore stated every premise, de-duplicated the theorem set into three groups, listed the missing large-scale experiments, and declined every superiority claim. A second risk is dismissing an entire architecture family from a negative result; NT01 and Section 4 argue specifically against that reading. Generative AI tools were used for implementation assistance and language editing; all research questions, conjectures and conclusions were specified by the human author, who takes full responsibility.

Appendix A Complete Proofs of NT01-NT15

All 15 proofs are analytic. The finite-identity checks in the supplementary artifact verify statement formation and non-empty premises only; they are not part of any proof. Symbol convention: t = time index, tau = lag, d = t - s.

A.1 NT01 - Tight minimum tensor order for exact single-hop retrieval

Premises. Static LTI kernel K_n(d)>0 for every delay d>=0 (strictly positive); history contains n>=2 pairs (k_p,v_p) with independently assignable one-hot values; q=k_{j*}; code rows satisfy <Code[a],Code[b]>=R*delta_{a,b}; read-out is z_n = sum_{p,r} w_{p,n} q_r s_{p,r,n,L} with s_{p,r,n,t}=rho_{p,n}s_{p,r,n,t-1}+a_t[r]b_t[n].

Claim. (a) A vector-valued (order-1) state that is a fixed-dimensional linear summary of the history cannot implement the retrieval, for ANY read-out. (b) An order-2 tensor state attains it EXACTLY, with zero interference and WITHOUT knowing the target position: z = R*K_n(L-tau_{j*})*v_{j*}, hence argmax_n z_n = argmax_n v_{j*,n} because K_n>0. Therefore the minimum tensor order is exactly 2, and the bound is TIGHT.

(a) Let A=L(v_1..v_n) in R^d be the summary, L: R^{nd}->R^d linear and independent of q. rank(L)<=d<nd for n>=2, so dim ker L>=nd-d>0. Pick nonzero Delta-v in ker L supported on the target coordinate: two value assignments give identical (A,q) hence identical f(A,q) for any f, but require different answers. Contradiction.

(b) Unroll the recurrence: s_{p,r,n,L}=sum_tau rho_{p,n}^{L-1-tau} a_tau[r] b_tau[n]. Contract with q_r and w_{p,n}: z_n = sum_tau (q . k_tau) K_n(L-1-tau) v_{tau,n} with K_n(d)=sum_p w_{p,n} rho_{p,n}^d. By code orthogonality q . k_tau = R delta_{tau,j*}, so every non-target term vanishes EXACTLY and z_n = R K_n(L-1-tau_{j*}) v_{j*,n}. Since rho_{p,n}>0 and w>0 give K_n>0, the positive scalar cannot change the argmax over n. No position information is needed at read-out. Hence order 2 suffices.

Tightness: (a) forbids order 1 and (b) achieves order 2. QED

Boundary. Requires K_n(d)>0; a damped-oscillator kernel sin((tau+1)theta) changes sign and is NOT admissible. Requires one-hot/independently assignable values and an exactly orthogonal code; NT08 gives the robust version. Classification/argmax retrieval, not real-valued vector reconstruction.

Architectural use. Explains why the general real-pole dictionary beats the oscillator grid, and turns 'TSC works' into a tight optimality statement rather than an empirical accuracy. *Source:* Section 4 TSC; core/arch/sgc.py general second-order IIR dictionary

A.2 NT02 - Strict-positivity separation of the two pole families

Premises. Second-order IIR y_t = a y_{t-1} + b y_{t-2} + c u_t with characteristic roots either (i) two positive reals 0<r2<=r1<1, a=r1+r2, b=-r1*r2, or (ii) a conjugate pair rho*e^{+-i theta}, 0<rho<1, 0<theta<pi, a=2 rho cos(theta), b=-rho^2.

Claim. Family (i) has h(tau)>0 for every tau>=0; family (ii) has h(tau)<=0 for some tau<=ceil(pi/theta). Hence the two impulse-response families are DISJOINT, and membership is decided exactly by inspecting the sign sequence of h(0..L-1) in O(L) (zero misclassification in exact arithmetic). Moreover the map (r1,r2)->(a,b) is injective on the ordered real-root domain, so the real-pole parameterization has no redundant columns.

(i) h(tau)=sum_{j=0}^{tau} r1^{tau-j} r2^{j} = (r1^{tau+1}-r2^{tau+1})/(r1-r2) for r1!=r2, and (tau+1)r^{tau} for r1=r2=r. Every term is strictly positive, so h(tau)>0.

(ii) h(tau)=rho^tau sin((tau+1)theta)/sin(theta) up to a positive normalization. Let tau0=ceil(pi/theta). Then (tau0+1)theta >= pi+theta > pi and (tau0+1)theta <= pi+2theta < 3pi. If pi < (tau0+1)theta < 2pi the sine is negative; if it reaches [2pi,3pi) take tau0-1 instead, for which (tau0)theta in [pi, pi+theta) subset [pi,2pi), so sin<=0. In all cases some tau<=ceil(pi/theta) gives sin((tau+1)theta)<=0, hence h(tau)<=0 (rho^tau>0, sin(theta)>0). A strictly positive sequence cannot contain a non-positive term, so the families are disjoint.

Injectivity: (a,b) determines the monic polynomial z^2 - a z - b, whose roots are exactly {r1,r2}; ordering r2<=r1 makes the pair unique. Real distinct roots require discriminant a^2+4b>0; both roots in (0,1) require 0<a<2 and -1<b<0. QED

Boundary. Root r1=r2 is a double real root (still strictly positive), handled by the limit formula. theta=pi is excluded by the architecture's theta clamp. Sign inspection needs exact arithmetic; quantization is covered by NT08.

Architectural use. Gives a provable, O(L)-checkable discriminant between the oscillator basis and the general second-order dictionary, and shows the latter has no parameter redundancy (unlike theta->0 collinearity, NT06). *Source:* core/arch/sgc.py _init_oscillator_grid / general second-order dictionary

A.3 NT03 - Provable separation of rho<1 multi-pole TSC from the linear-at

Premises. TSC read-out kernel K_n(d)=sum_{p=1}^{P} w_{p,n} rho_{p,n}^{d} with P>=1, 0<rho_{p,n}<1, w_{p,n}>0, all rho_{p,n} distinct. Linear-attention (single-pole, rho=1) kernel is the constant K(d)=1.

Claim. The span of {K_n : n} contains NO nonzero constant function, and the constant function is not in the closure of that span in l^infty(N). Hence the rho<1 multi-pole kernel family is STRICTLY SMALLER than the rho=1 single-pole family as a set of decay profiles, and the two state-update operators are not equal. In particular the TSC state is not the linear-attention state S=sum_tau k_tau v_tau^T once rho<1.

Any nonzero combination F(d)=sum_n c_n K_n(d) is a finite sum of exponentials with bases in (0,1). Let rho_max<1 be the largest base; then F(d)=Theta(rho_max^d) -> 0 as d->infinity. A constant function equal to 1 does not tend to 0, so no finite combination equals it. For the l^infty closure: ||F - 1||_infty >= sup_{d>=D} |1-F(d)| -> 1 for every F in the span, so the distance from the span to the constant is exactly 1 >0 and the constant is not a limit point either.

Consequently the rho=1 update s_t=s_{t-1}+u_t (constant kernel, state S=sum k_tau v_tau^T) and the rho<1 update s_t=rho s_{t-1}+u_t (decaying kernel) generate different linear operators on the same tensor signal; they cannot be identified by any change of basis that preserves the impulse response. The rho->1, P=1 limit recovers the isomorphism, which is therefore a boundary case, not the general case. QED

Boundary. This is a separation of KERNEL FAMILIES / state-update operators, not a claim that the full models have different Turing-complete expressivity, nor a claim of empirical superiority. It does not assert that TSC is outside every linear-attention VARIANT (e.g. data-dependent gates); it settles the specific rho<1 multi-pole vs rho=1 single-pole question.

Architectural use. Directly answers the reviewer's point 3: instead of leaving the TSC / linear-attention relation 'to the reader', we give a provable non-equivalence while still recording the rho->1 isomorphism. *Source:* Section 4 remark (rho->1 isomorphism) + core/theory/smm_sgc.py

A.4 NT04 - Deterministic zero-crosstalk codes for arbitrary V are 2-optima

Premises. V>=1 arbitrary integer; code rows in R^R with zero crosstalk.

Claim. (a) Zero crosstalk among V nonzero rows forces R>=V. (b) R=2^ceil(log2 V), Code_V = first V rows of the Sylvester Hadamard H_R, satisfies Code_V Code_V^T = R I_V exactly and R<2V. Hence the construction attains approximation ratio strictly below 2 against the R>=V lower bound, and no Hadamard-prefix construction can do better than the next power of two.

(a) V mutually orthogonal nonzero vectors in R^R are linearly independent, so V<=R.

(b) Induction on the Sylvester recursion H_{2m}=[[H_m,H_m],[H_m,-H_m]] gives H_R H_R^T = R I_R; the leading V-by-V principal submatrix of that identity is exactly Code_V Code_V^T = R I_V. Since R=2^ceil(log2 V) < 2^{log2 V + 1} = 2V for V>1 (and R=V=1 for V=1), R/V < 2. Any Hadamard-prefix construction must use an order that is a power of two and at least V, so R is the minimum such order; the ratio is therefore optimal within this deterministic class. QED

Boundary. Existence + explicit construction + optimality within Hadamard prefixes. Does NOT prove deployed-kernel cheapness or that a non-Hadamard code cannot reach R=V (e.g. an orthonormal V-by-V basis would, but is not a deterministic +/-1 code).

Architectural use. Closes the V>>64 construction question with a clean 2-approximate bound. *Source:* CT5-L2 / core/theory/smm_sgc.py DETERMINISTIC_CODE

A.5 NT05 - State-vocabulary lower bound and end-to-end tightness

Premises. Static LTI tensor state s_{p,r,n} with P pole channels, R code rows, N value channels; exact zero-crosstalk retrieval of V token classes under NT01's premises.

Claim. Any such construction requires at least P*V*N scalar state registers, i.e. Omega(V*N). TSC with the NT04 code uses P*R*N < 2 P*V*N registers, so it attains the lower bound within a factor 2 END-TO-END: code dimension, state size and retrieval exactness are simultaneously tight.

Zero crosstalk requires R>=V mutually orthogonal code rows (NT04a). The state is indexed by (p,r,n); for each of the P*N pairs (p,n) the R code coordinates must be stored independently because the recurrence s_{p,r,n,t}=rho_{p,n}s_{p,r,n,t-1}+a_t[r]b_t[n] is diagonal in r and no linear combination of the r-coordinates is invariant under all one-hot inputs a_t=Code[x]. Formally, the observability matrix of the R parallel first-order systems has full column rank R, so a faithful state needs >=R registers per (p,n). Hence total >=P*R*N>=P*V*N. With R<2V from NT04b the construction uses <2P*V*N: ratio <2. NT01 supplies exactness, NT04 supplies the code, so all three bounds are met simultaneously. QED

Boundary. Lower bound is for faithful (injective-in-r) static LTI tensor states with a fixed code. Compressed or learned non-tensor states are outside the premise. The factor 2 is a guarantee, not optimality of the constant.

Architectural use. Turns the TSC construction into a tight state-complexity result instead of a single accuracy number on V=64. *Source:* Section 4 cost remark (56,320 B state) + NT01/NT04

A.6 NT06 - Quantitative rank-deficiency rate of near-zero frequency grids

Premises. Dictionary columns f_m(tau)=rho^tau sin((tau+1)theta_m), 0<rho<1, theta_1<theta_2<...<theta_M, Delta=theta_2-theta_1, Gram matrix G_{ij}=<f_i,f_j> over tau>=0.

Claim. cond(G) >= 2*sin^2(theta_1) / (Delta^2 * A(rho)), where A(rho)=sum_{tau>=0} rho^{2tau}(tau+1)^2 = (1+rho^2)/(1-rho^2)^3 < infinity. Thus a grid whose first spacing satisfies Delta << theta_1 forces cond(G) = Omega((theta_1/Delta)^2); a grid with Delta ~ theta_1/M and theta_1 = Theta(1) keeps the bound at O(M^2) instead of exponential.

Use the Rayleigh quotient with the vector e_1-e_2: sigma_min(G) <= (1/2)||f_1-f_2||^2. By |sin a - sin b| <= |a-b|, ||f_1-f_2||^2 = sum_tau rho^{2tau}[sin((tau+1)theta_1)-sin((tau+1)theta_2)]^2 <= Delta^2 sum_tau rho^{2tau}(tau+1)^2 = Delta^2 A(rho). The closed form follows from sum n^2 x^n = x(1+x)/(1-x)^3 with x=rho^2 and index shift. Also sigma_max(G) >= G_{11} = ||f_1||^2 >= rho^0 sin^2(theta_1) = sin^2(theta_1). Hence cond = sigma_max/sigma_min >= 2 sin^2(theta_1)/(Delta^2 A(rho)). For logfreq grids with theta_1 extremely small and Delta proportional to theta_1*(r-1), the two small quantities nearly cancel in the RATIO but A(rho) explodes as rho->1; for linear grids theta_1=Theta(1) and Delta=Theta(1/M), giving cond >= Theta(1/M^2 * 1/A(rho)) while the measured value is O(1), i.e. the bound is not tight there but correctly predicts that near-zero theta_1 combined with tiny Delta destroys conditioning. QED

Boundary. A LOWER bound on cond (a badness certificate), not an exact value and not an upper bound. The linear-grid O(1) measurement is consistent with a loose lower bound; the theorem explains the FAILURE of near-zero grids, not the full success of linear ones. Condition numbers remain observations.

Architectural use. Upgrades the earlier 'direction only' asymptotic into a checkable rate, and explains why theta_min=pi/8192 collapses capacity while a linear grid restores it. *Source:* Prop8 / docs/frequency_grid_fix.md / core/arch/sgc.py init_mode

A.7 NT07 - Precision-length-decay trilemma with an interior optimal rho

Premises. First-order streaming state s_t=rho s_{t-1}+u_t, 0<rho<1, |u_t|<=U; each update is quantized with absolute step error <= delta, delta/U < 1; a distant contribution at delay L has amplitude rho^{L-1} U and must exceed c times the accumulated error floor (c>1 a fixed separation constant); c*delta/U < 1.

Claim. (a) The accumulated error is bounded by e_infty <= delta/(1-rho). (b) The resolvable length obeys L(rho) <= 1 + ln((1-rho)U/(c*delta)) / |ln rho|. (c) L(rho) is continuous on the open interval (0, 1-c*delta/U), tends to 0 at BOTH endpoints, and is positive inside; hence it attains an INTERIOR MAXIMUM at some rho* strictly between 0 and 1. Therefore rho->1 is NOT optimal: longer memory requires an interior decay, and rho=1 admits no resolvable length at all.

(a) The error obeys e_t <= rho e_{t-1} + delta; by induction e_t <= delta sum_{k=0}^{t-1} rho^k < delta/(1-rho).

(b) Resolvability requires rho^{L-1} U > c*delta/(1-rho), i.e. (L-1) ln rho > ln(c*delta/((1-rho)U)). Dividing by ln rho<0 reverses the inequality and gives L-1 < ln((1-rho)U/(c*delta))/|ln rho| =: g(rho).

(c) g requires (1-rho)U > c*delta, i.e. rho < 1 - c*delta/U. As rho -> (1-c*delta/U)^- the numerator -> 0^+ and the denominator is bounded away from 0, so g -> 0^+. As rho -> 0^+, |ln rho| -> infinity while the numerator tends to the finite positive ln(U/(c*delta)), so g -> 0^+. g is continuous and strictly positive somewhere inside (e.g. at rho=1/2 when delta/U is small), so by the extreme value theorem on a compact subinterval it attains a positive interior maximum. At rho=1 the error floor delta/(1-rho) is unbounded, so no L satisfies (b). QED

Boundary. CONDITIONAL on a uniform absolute quantization-error model (same premise as the perturbation bound). g is a SUFFICIENT resolvable-length bound, not an exact measured drop point; dynamic per-step scaling changes the effective delta, so O6's measured L=16385 is not derived from it. The existence of rho* is proved, its numerical value is an observation.

Architectural use. The flagship quantitative result: proves that the best decay is interior, explains rho=0.999 as a principled compromise, and turns 'rho is the only lever' from a measurement into a theorem with a rate. *Source:* O6 physics probes + Appendix A.7 perturbation bound + T12/T13

A.8 NT08 - Robust exact retrieval under bounded code perturbation

Premises. NT01's architecture, but the code satisfies Code Code^T = R I_V + E with ||E||_infinity <= epsilon; all kernels K_n(d) in [K_min,K_max] with K_min>0; values are one-hot with |v|<=1; at most L-1 non-target terms.

Claim. The retrieval remains exactly correct (argmax unchanged) whenever epsilon < R*K_min / ((L-1)*K_max). More generally the non-target interference is bounded by epsilon*(L-1)*K_max, so the target margin is at least R*K_min - epsilon*(L-1)*K_max.

Write q.k_tau = R delta_{tau,j*} + E_{q,tau}. Then z_n = R K_n(L-1-tau_{j*}) v_{j*,n} + sum_{tau != j*} E_{q,tau} K_n(L-1-tau) v_{tau,n}. The target coordinate of the one-hot value has amplitude >= R K_min. Each non-target term is bounded by epsilon*K_max*1 and there are at most L-1 of them, so the total interference on any coordinate is <= epsilon (L-1) K_max. If that is strictly less than R K_min, the target coordinate strictly dominates every other coordinate and argmax is unchanged. The stated inequality is exactly this condition. QED

Boundary. A SUFFICIENT condition, not necessary; assumes bounded kernels and one-hot values. It covers quantized codes, learned approximate codes and random orthogonal projections, explaining why O5's random-code arm also reached 1.0000 without any retraining.

Architectural use. Makes NT01 deployable: the deterministic Hadamard code is not a fragile requirement, only a bounded-crosstalk one. *Source:* Section 4 code remark + O5 random orthogonal code control

A.9 NT09 - Rounding-order crossover between the FFT and IIR paths

Premises. Standard floating-point model with unit roundoff eps; an L-point radix-2 FFT has log2(L) butterfly stages with relative error at most c1*eps per stage; the stable second-order IIR has steady-state relative error at most c2*eps/(1-rho), independent of L; c1,c2>0 fixed.

Claim. The IIR error bound is L-independent while the FFT bound grows as c1*eps*log2(L). They cross at L* = 2^{c2/(c1(1-rho))}. For every L < L* the streaming path has the smaller worst-case bound; for rho close to 1, L* is astronomically large, so on all realizable sequence lengths the IIR

bound is never worse.

FFT: after log2 L stages, (1+c1 eps)^{log2 L} - 1 <= c1 eps log2 L + O(eps^2). IIR: the perturbation recursion e_t <= rho e_{t-1} + c2 eps has the steady state e <= c2 eps/(1-rho), with no L dependence because the filter is BIBO stable and the bound is the l1 norm of its impulse response. Equating c1 eps log2 L = c2 eps/(1-rho) cancels eps and gives log2 L* = c2/(c1(1-rho)), i.e. L* = 2^{c2/(c1(1-rho))}. Monotonicity of log2 L and constancy of the IIR bound give the two regimes. QED

Boundary. CONDITIONAL on the stated floating-point model and on WORST-CASE BOUNDS, not measured error. It does not contradict the withdrawal of unconditional fp32 bitwise duality (NT/Q02): equality is still only up to tolerance. It is a selection criterion between execution orders, not a speed claim.

Architectural use. Supplies the missing numerical-selection rule for the dual-form theorem: which path to trust at a given L and rho. *Source:* Thm1 remark + backend 16-op contract + Appendix A.7

A.10 NT10 - Per-channel time-invariance preserves the duality

Premises. TSC recurrence s_{p,r,n,t}=rho_{p,n} s_{p,r,n,t-1}+u_t[r,n] with rho_{p,n} depending on the pole channel p and value channel n but NOT on time t or code index r; coefficient tensor w_{p,n} likewise static.

Claim. Each (p,n) pair is an independent LTI system with transfer function H_{p,n}(z)=1/(1-rho_{p,n} z^{-1}). The full operator is BLOCK Toeplitz (Toeplitz in t for each channel, diagonal in (p,n)), so a single batched FFT diagonalizes it channel-wise and the streaming recurrence realizes the same operator. The state has P*R*N scalars, independent of L. Consequently a per-channel coefficient tensor of shape (M,D) is architecturally admissible without breaking any purity constraint.

For fixed (p,r,n), unrolling gives s_{p,r,n,t}=sum_tau rho_{p,n}^{tau} u_{t-tau}[r,n], whose impulse response depends only on the delay tau. Stacking the P*N independent channels gives a block-diagonal-in-(p,n) operator whose t-blocks are Toeplitz; by NT12 (applied per channel) the parameter must be constant in t, which it is. A Toeplitz operator is diagonalized by one DFT per channel, so one batched FFT suffices; the recurrence is the same rational function computed in the time domain. The state count is P*R*N with no L term. QED

Boundary. Requires rho and w independent of t. It does NOT permit input-dependent rho_t (that is the SFSC/LTV case, excluded by NT12/Q09). The constant P*R*N may be large (56,320 B observed); O(1) is asymptotic.

Architectural use. Upgrades the v4 finding (coef was artificially flattened from (32,2048) to 32 scalars) from a script simplification into a theorem about what the architecture permits. *Source:* core/arch/sgc.py coef (M,D) + T7 / Section 4

A.11 NT11 - Layer-count lower bound for multi-hop retrieval

Premises. A stack of L layers, each a polynomial map of total degree at most d in its inputs (a bilinear contraction has d=2); skip connections do not raise degree; the target H-hop output contains H key-match factors and one value factor, i.e. total degree >= H+1 in the history variables.

Claim. Any such stack needs L >= ceil(log_d(H+1)) layers. For bilinear layers L >= ceil(log2(H+1)); in particular H=2 already requires L>=2, so a single contraction layer cannot implement two-hop retrieval. This is a NECESSARY condition; no sufficient construction is asserted.

The composition of polynomials of degrees d_1..d_L has degree at most the product d_1...d_L <= d^L (degree cannot increase under addition or skip connections). The H-hop target is a multilinear form of degree H+1 in (k,v): H factors of the form <phi(q),psi(k)> plus one value factor. A polynomial of degree < H+1 cannot equal a nonzero degree-(H+1) form on an open set. Hence d^L >= H+1, i.e. L >= log_d(H+1); taking ceil gives the integer bound. For d=2, H=2: log2(3)=1.58 -> L>=2. QED

Boundary. NECESSARY only. The degree bound is an upper bound on what L layers CAN reach; reaching it needs a construction (untested). Per-position nonlinearities of degree >2 change d. The measured H4_multihop=0.0322 is a finite-sample observation, not a theorem conclusion.

Architectural use. Gives the stacking roadmap a concrete quantitative target instead of 'try more layers', and explains the two-hop collapse structurally. *Source:* T15 / L7 / O5 H4_multihop; core/arch stacking

A.12 NT12 - Toeplitz iff time-invariant coefficients, for arbitrary order and

Premises. y_t = sum_{j=1..P} a_{j,t} y_{t-j} + c_t u_t with c_t != 0 for all t; P>=1 arbitrary; optionally with an independent copy for each of C channels (a_{j,t}^{(c)}, c_t^{(c)}).

Claim. The impulse response h(t,s) depends only on t-s IFF every coefficient a_{j,t} and c_t is constant in t. With C independent channels the full operator is block Toeplitz IFF this holds for every channel. Necessity fails at the first delay where c_t=0, so the theta-clamp / nonzero-c premise is essential, not decorative.

Necessity by induction on the delay k. k=0: h(s,s)=c_s must be s-independent, so c_t=c. Assume a_1..a_{k-1} are constant. At delay k, h(s+k,s) = a_{k,s+k} c + Phi(a_1..a_{k-1},c), where Phi is a polynomial with s-independent coefficients. Toeplitz forces h(s+k,s) independent of s, and c != 0 gives a_{k,t} constant. Induct to k=P. Sufficiency: constant coefficients give the LTI transfer function H(z)=c/(1-sum_j a_j z^{-j}), so h(t,s)=h(t-s). For C channels apply the same argument to each diagonal block; off-diagonal blocks are zero. If c_t=0 for some t, the k=1 step cannot isolate a_{1,t}, and a counterexample with time-varying a_1 but Toeplitz h exists (the input never enters at that step). QED

Boundary. The equivalence concerns THIS convolution/FFT path; it does not assert nonexistence of other sub-quadratic algorithms. c_t != 0 is guaranteed in SGC by sin(theta)>0 under the theta clamp; removing the clamp invalidates the necessity direction.

Architectural use. The precise equivalence-strengthening the reviewer asked to foreground; now covers arbitrary order and per-channel grids, and identifies exactly where the premise can break. *Source:* Thm4 + T11 + Q09/Q11; core/arch/sgc.py theta_clamp_floor

A.13 NT13 - Arbitrary-read-out impossibility with a sharp quantitative bou

Premises. EXACTLY four premises: (i) the history enters the final computation only through a fixed-dimensional LINEAR summary A=L(v_1..v_n) in R^d that is independent of q; (ii) q enters only at the current position; (iii) values are independently assignable; (iv) at least two distinguishable targets exist. Read-out f(A,q) is arbitrary (linear or nonlinear). Key linear independence is NOT required.

Claim. (a) Content addressing is impossible for any f. (b) Quantitatively dim ker L >= nd - d. (c) For n=1 the kernel can be zero, so premise (iv) is NECESSARY and cannot be dropped. (d) TSC escapes by changing the tensor order of the signal and the read-out, NOT by contradicting (a).

(a,b) rank(L)<=d, so dim ker L = nd - rank(L) >= nd-d > 0 for n>=2. Choose nonzero Delta-v in ker L supported on a coordinate of the target value (possible by (iii),(iv)). Two assignments v and v+Delta-v give the same A and the same q, hence the same f(A,q), but require different answers. Contradiction for every f. Linear read-out follows by subtracting two queries. (c) n=1 gives nd=d; a random square L is full rank almost surely, so ker=0 and (a) no longer follows -- the premise is sharp. (d) NT01 uses an order-2 tensor state, which is not a fixed vector summary, so (i) is false; the theorem is inapplicable, not violated. QED

Boundary. DOES NOT extend to deep stacks with per-position nonlinearities that mix A and q, to input-dependent kernels (NT12 instead), or to outer-product states (NT01). Corollary B1.1 ('requires a pairwise term') is an EXPLANATORY FRAMEWORK, NOT A THEOREM, and is not a proved sufficient or necessary characterization of all escape mechanisms.

Architectural use. The strongest boundary theorem, with an explicit four-premise contract that must appear in the abstract and conclusion to prevent misreading. *Source:* Thm5 + T10 + CT4; Appendix A.5

A.14 NT14 - Information-Gradient Duality (with the softening trap)

Premises. An architecture has (a) an online state s_L=F(x_{1:L}) in R^D with D independent of L and F continuous; (b) an episodic bank of C EXACT copies read by hard top-k selection S_k(q) followed by continuous softmax weights; (c) no N x N score matrix, no resident KV cache; (d) the task requires exact retrieval of an arbitrary one of L history items, |X|>1, L unbounded.

Claim. No single channel is simultaneously (I) differentiable, (II) covering all L items, and (III) losslessly retrieving any one. The three horns are exclusive AND the natural repair is self-defeating: replacing hard top-k by a differentiable soft aggregate over all C copies re-triggers NT13's kernel impossibility.

(i) Differentiable + covering forces lossy: F must be injective on X^L, but a B-byte state has at most 2^{8B} configurations and |X|^L > 2^{8B} for L > 8B/log2|X| (pigeonhole).

(ii) Hard top-k: on the tie-free open set the ordering is locally constant, so d 1[j in S_k]/d theta = 0 a.e. and the Jacobian support is contained in S_k of size k; the C-k unselected items contribute exactly zero to any gradient update. Exact on k slots, fails (I) for selection and (II) because k<L.

(iii) SOFTENING: m_soft(q)=sum_{j=1..C} alpha_j(q) v_j is differentiable and covers all C, but for fixed q the map (v_1..v_C) in R^{Cd} -> m_soft in R^d is LINEAR with rank <= d, so dim ker >= Cd-d > 0 for C>=2. A nonzero Delta-V in that kernel leaves m_soft and q unchanged while changing the target value, so any f(m_soft,q) gives identical output for distinct required answers. This is exactly NT13. Hence (I)+(II) again forces NOT (III).

The three repairs are mutually exclusive. QED

Boundary. Does not forbid hard top-k retrieval of k items (the L2 pointer sidecar), nor a zero-order path (NT15). It says nothing about empirical LM quality. Numerical checks are finite identities, not the proof.

Architectural use. The hardest structural theorem: turns 'gradient vanished' into a closed-loop impossibility with a self-defeating repair. *Source:* CT6 / core/theory/smm_sgc.py; memory.py topk, smm.py @torch.no_grad

A.15 NT15 - Zero-order self-evolution: gate validity and information rate

Premises. NT14 holds; online answer path has no autograd graph; consolidation writes to weights only through a no-regression gate comparing paired-bootstrap one-sided CI lower bounds l_alpha (level 1-alpha) against -|tau| over m metrics, rejecting if ANY fails; an independent held-out verifier is supplied (otherwise writeback is refused); at most w insights of b bits are committed per round; each read selects k items.

Claim. (a) P(false commit | true regression) <= alpha for one metric and <= m*alpha for m metrics (<= alpha under Bonferroni alpha/m). (b) I(history ; weight increments over T rounds) <= T*w*b bits, INDEPENDENT of L and C; instantaneous bandwidth <= k*b bits/round. (c) The path MUST be zero-order. (d) A non-decreasing ACCEPTED score does NOT imply monotone real capability.

(c) NT14(ii) zero Jacobian for unselected items, NT14(iii) softening destroys exactness, and the no_grad online path has no graph: no gradient can carry long-context experience into the weights, so the only admissible channel is a zero-order accept/reject test on verifier outputs.

(a) Coverage: P(l_alpha <= Delta_true) >= 1-alpha, hence P(l_alpha > Delta_true) <= alpha. A false commit requires l_alpha >= -|tau| > Delta_true, i.e. l_alpha > Delta_true, exactly the coverage-failure event. Union bound over m metrics gives m*alpha.

(b) By data processing, I(H;Delta theta_t) <= H(insights_t, verifier_t) <= O(w*b). Summing over T rounds gives T*O(w*b). L and C do not appear because NT14(ii) bounds each read to k selected items; the instantaneous bound is k*b.

(d) The gate proves NO-REGRESSION, a necessary condition for writeback; improvement requires a separate held-out evaluation. The upstream docstring states exactly this. QED

Boundary. (a) CONDITIONAL on paired i.i.d. samples, bootstrap consistency and a non-default verifier; with the default constant verifier the loop refuses writeback, so the gate is inactive, not contradicted. (b) is an UPPER BOUND, not a claim the loop reaches it. Monte-Carlo checks are ILLUSTRATION ONLY. T16's scalar-monotonicity claim is WITHDRAWN (W7).

Architectural use. First rate bound for the SMM x SGC self-evolution path; shows long context buys inference-time retrieval, NOT a faster learning rate. *Source:* CT7 / dream/evaluation.py regression_gate, loop.py VERIFY, hooks.py

A.16 NT16 - Exact Turan identity and log-convexity of the real-pole diction

Premises. General second-order IIR y_t = a y_{t-1} + b y_{t-2} + u_t with characteristic roots 0 < r2 <= r1 < 1, so a = r1+r2, b = -r1*r2; impulse response h(tau) with h(0)=1, h(-1)=0.

Claim. (a) EXACT IDENTITY: h(tau)^2 - h(tau-1) h(tau+1) = (r1*r2)^tau > 0 for every tau >= 1, including the double-root case r1=r2=r where it becomes r^{2tau}. Hence h is STRICTLY log-convex. (b) The conjugate-pole kernel h(tau) = rho^tau sin((tau+1)theta)/sin(theta) is NOT log-convex: it takes a negative value at some finite tau, whereas a log-convex sequence with h(0)>0 is strictly positive for all tau. (c) Therefore membership in the real-pole family is equivalent to log-convexity of the impulse response, a SUFFICIENT statistic that needs no root solving.

(a) Binet form: h(tau) = (r1^{tau+1} - r2^{tau+1})/(r1-r2) for r1>r2, and h(tau) = (tau+1) r^tau for r1=r2=r. For r1>r2 set A=r1^{tau}, B=r2^{tau}, p=r1, q=r2 and expand h(tau)^2 - h(tau-1)h(tau+1). Writing s_k = (p^{k+1}-q^{k+1})/(p-q) and using s_{n+1} = (p+q)s_n - pq s_{n-1}, the Hankel determinant D_tau = s_tau^2 - s_{tau-1}s_{tau+1} obeys D_{tau+1} = pq D_tau with D_1 = pq, hence D_tau = (pq)^tau = (r1 r2)^tau > 0. The double-root case follows by continuity (or by the identity (tau+1)^2 - tau(tau+2) = 1), giving r^{2tau}. Since D_tau>0 and h(0)=1>0, induction gives h(tau+1) > h(tau)^2/h(tau-1) > 0, so h is strictly positive and strictly log-convex.

(b) By NT02 there exists tau0 <= ceil(pi/theta) with sin((tau0+1)theta) <= 0, hence h(tau0) <= 0. A log-convex sequence with h(0)>0 satisfies h(tau)>0 for all tau by the same induction, contradiction.

(c) The two families are exhaustive for stable second-order roots (discriminant sign), and (a)/(b) separate them by a checkable inequality on the impulse response alone. QED

Boundary. Normalization h(0)=1 (c=1) is assumed; a general c>0 scales all h by c and leaves D_tau = c^2 (r1r2)^tau > 0, so log-convexity is invariant. theta=pi excluded by the architecture's clamp. The Turan identity is exact over R; finite precision is NT08/NT25.

Architectural use. Gives the STRUCTURAL reason the general second-order dictionary beats the damped-oscillator grid on content addressing (NT01 needs K_n>0), and turns NT02's O(L) sign scan into a log-convexity certificate. It explains why the 'oscillatory basis' was the wrong inductive bias. *Source:* core/arch/sgc.py general second-order IIR dictionary + NT02

A.17 NT17 - O(1) three-sample identifiability and pole-family discriminant

Premises. A stable second-order recursion with impulse response normalized to h(0)=1; the first three samples h(0),h(1),h(2) are observed exactly.

Claim. (a) The recursion coefficients are recovered EXACTLY and uniquely: a = h(1)/h(0), b = (h(2) - a h(1))/h(0). (b) The 2x2 Hankel determinant det H_2 = h(0)h(2) - h(1)^2 = b != 0, so the minimal linear realization has rank exactly 2. (c) The discriminant Delta = a^2 + 4b computed from these three samples satisfies Delta = (r1-r2)^2 >= 0 for real poles and Delta = -4 rho^2 sin^2(theta) < 0 for conjugate poles, with equality Delta=0 iff the real roots coincide. Hence the pole family is decided in CONSTANT time from three samples, strictly stronger than NT02's O(L) sign scan and independent of L.

(a) From y_t = a y_{t-1} + b y_{t-2} + u_t with the unit impulse, h(0)=1, h(1)=a h(0)=a, h(2)=a h(1)+b h(0)=a^2+b. Solving gives the stated inversion; uniqueness follows because (h(1),h(2)) determines (a,b) by a triangular system.

(b) det H_2 = h(0)(a^2+b) - a^2 = b. Real roots give b=-r1r2<0; conjugate roots give b=-rho^2<0; both nonzero, so rank(H_2)=2.

(c) The characteristic polynomial is z^2 - a z - b with roots (a +- sqrt(a^2+4b))/2. For real roots r1,r2 we have a=r1+r2 and b=-r1r2, so a^2+4b=(r1+r2)^2-4r1r2=(r1-r2)^2>=0. For a=2rho cos(theta), b=-rho^2 we get a^2+4b = 4 rho^2(cos^2 theta - 1) = -4 rho^2 sin^2 theta < 0 for theta in (0,pi). Equality Delta=0 forces r1=r2 (real) or sin theta=0 (excluded). Since (a,b) come from three samples, Delta does too, with no dependence on L. QED

Boundary. Requires exact h(0..2); with noise the discriminant is a

finite-sample estimate (NT08 gives the robust version). Rank 2 is the MINIMAL LINEAR realization rank, not a claim about nonlinear realizations.

Architectural use. An O(1) architecture diagnostic: before training, three impulse samples tell whether a grid is real-pole or conjugate-pole and whether its roots coincide (collinearity risk, NT06). Useful for init_mode validation in sgc.py. *Source:* core/arch/sgc.py _init_oscillator_grid / build_kernel

A.18 NT18 - Linear grids maximize the minimum spacing under a theta_min budget

Premises. M frequencies theta_1 < ... < theta_M in [theta_min, theta_max], theta_min > 0 fixed; let Delta_min = min_i (theta_{i+1}-theta_i). A grid is admissible if it respects the interval.

Claim. (a) Delta_min <= (theta_max - theta_min)/(M-1), with EQUALITY iff the grid is exactly equally spaced (linear). (b) Consequently, among all admissible grids, the linear grid uniquely maximizes the smallest pairwise gap, and therefore uniquely maximizes the NT06 conditioning lower bound among grids sharing (theta_min, theta_max, M). (c) Any logfreq grid whose local gaps near theta_min shrink geometrically has Delta_min <= theta_min(r-1), so its NT06 bound is strictly worse than linear's by a factor that grows as theta_min -> 0. A hybrid grid is a convex interpolation and is bounded between them.

(a) Summing the M-1 gaps gives theta_M - theta_1 <= theta_max - theta_min. Hence (M-1) Delta_min <= sum gaps <= theta_max - theta_min, giving the bound. Equality requires every gap equal AND theta_1=theta_min, theta_M=theta_max, i.e. the linear grid.

(b) NT06 gives cond >= 2 sin^2(theta_1)/(Delta^2 A(rho)); with theta_1 and A(rho) fixed by the budget, maximizing the bound means maximizing Delta_min, which (a) shows is the linear grid, uniquely.

(c) For a logfreq grid theta_{i+1}=r theta_i, the smallest gap is theta_1(r-1) <= theta_min(r-1) (equality at the first gap). A hybrid grid is a mixture theta_i = alpha*linear_i + (1-alpha)*logfreq_i; its minimum gap is at least alpha*Delta_linear + (1-alpha)*Delta_logfreq and at most Delta_linear, so its conditioning bound lies between the two. QED

Boundary. Optimizes the MINIMUM gap / the NT06 LOWER bound, not the exact condition number (which is a measured value). It does not claim linear is best for every task; NT06's bound is loose at large theta_1. Theta ranges and M are fixed by the budget.

Architectural use. Turns 'linear30 was best in a 3-boundary sweep' from a measurement into a uniqueness theorem, and explains quantitatively why hybrid sits between legacy and linear (docs/frequency_grid_fix.md). *Source:* docs/frequency_grid_fix.md + core/arch/sgc.py init_mode

A.19 NT19 - The two-tap state is a MINIMAL sufficient statistic (tight byte bound)

Premises. Stable second-order IIR with distinct OR repeated roots, per basis function m; D channels; M basis functions; streaming state stores the previous two outputs per (m,d).

Claim. (a) No deterministic state of fewer than 2 scalars per (m,d) can reproduce the same output sequence for all inputs. (b) Hence 2*M*D scalar registers are NECESSARY, not just sufficient, and the architecture's state expression 4(2MD + (K-1)D) bytes is tight up to the gate registers. (c) The O(1) state property (Thm3) is therefore OPTIMAL within causal LTI realizations: the constant cannot be reduced without changing the kernel family.

(a) Suppose a 1-scalar causal linear state s_t = lambda s_{t-1}+u_t realizes the operator. Its impulse response is h(tau)=lambda^tau, so h(1)^2 = h(0)h(2). But NT17 gives h(0)h(2)-h(1)^2 = -b != 0 for BOTH root families (b=-r1r2 or b=-rho^2). Contradiction. A nonlinear 1-scalar state cannot realize a linear operator whose Hankel rank is 2 (rank is realization-independent), so 2 is a lower bound for any CAUSAL LINEAR state realization. (Nonlinear states of unbounded precision are outside this premise; the architecture uses fixed-width fp32 registers.) The recurrence itself achieves 2, so the minimum is exactly 2.

(b) M basis functions and D channels are independent (block diagonal in (m,d)), so the minima add: 2MD. Adding the K-1 gate scalars per channel gives the source expression; the L-independent part is exactly the necessary part.

(c) Any causal LTI realization of a degree-2 rational transfer function has McMillan degree 2; by (a) this is a hard lower bound, so no re-parameterization reduces the constant. QED

Boundary. Minimality is within CAUSAL LINEAR state realizations of the SAME kernel (the family used by sgc.py); unbounded-precision nonlinear registers are outside the premise. Changing the kernel family (e.g. first-order rho-only) changes the premise. Gate registers are architecture-specific overhead, not part of the tight bound.

Architectural use. Answers the reviewer's 'state constant is large' concern with a tight lower bound: the 56,320 B TSC state is a DIFFERENT (tensor) construction, while the original mixer's 512 B cannot be shrunk without losing the second-order kernel. *Source:* core/arch/sgc.py streaming_state_bytes + Thm3 + NT17

A.20 NT20 - Axiomatic uniqueness of the no-regression min-rule gate

Premises. A consolidation gate maps m per-metric paired-bootstrap CI lower bounds (ell_1..ell_m) and a tolerance tau to {accept,reject}. Axioms: (P1) monotone: raising any ell_i cannot turn accept into reject; (P2) no compensation: if some ell_i < -|tau| then reject regardless of the others; (P3) permutation invariance; (P4) threshold invariance: the decision depends only on the bits a_i = 1[ell_i >= -|tau|]; (P5) non-triviality: a_i = 1 for all i implies accept.

Claim. The UNIQUE rule satisfying P1-P5 is G = AND_i [ell_i >= -|tau|], equivalently min_i ell_i >= -|tau|, which is exactly the upstream regression_gate. Consequently ANY weighted-sum, averaged, or majority-vote gate violates at least one axiom and permits a metric to collapse while still being compensated -- contradicting the source's 'no-regression' semantics.

By P4 the rule is a Boolean function g(a_1..a_m). By P2, any vector with some a_i=0 maps to reject. By P5 the all-ones vector maps to accept. These two facts determine g on all 2^m inputs: g = AND_i a_i. P1 and P3 are then automatically satisfied (AND is monotone and symmetric), so the rule exists and is unique. For a weighted sum S=sum_i c_i ell_i with c_i>0, choose ell_1 = -|tau|-epsilon and ell_2 = ... = ell_m = M large enough that S >= -|tau|: P2 is violated (a metric regressed but the round is accepted). Averaging and majority voting are special cases of such compensating rules. Hence they are excluded. QED

Boundary. Axioms P1-P5 are the formalization of the upstream docstring's semantics ('any metric fails -> reject'); the theorem is conditional on accepting that formalization. It concerns the AGGREGATION rule, not the statistical validity of the CI itself (that is NT15(a)).

Architectural use. Upgrades the gate from 'an engineering choice' to a forced characterization, and proves that the upstream implementation is the only one consistent with its own stated semantics. *Source:* simplcore/core/memory/dream/evaluation.py regression_gate

A.21 NT21 - The constant-verifier trap and necessity of an independent eva

Premises. The consolidation loop uses a verifier V(model) returning per-sample scores; the gate compares paired bootstrap CI lower bounds of new minus baseline scores; the loop refuses weight writeback when the verifier is the upstream DEFAULT (a constant function).

Claim. (a) If V is constant, every paired difference is exactly 0, every bootstrap resample has mean 0, the CI lower bound is 0 for every metric, and the gate therefore accepts EVERY round unconditionally: the gate carries ZERO information. (b) Hence 'refuse writeback when the default verifier is used' is not a conservative heuristic but a LOGICAL NECESSITY: it is the only way to keep the no-regression claim non-vacuous. (c) A non-constant, held-out verifier is NECESSARY and SUFFICIENT for the gate to be non-trivial (it can then reject).

(a) V(model)=c for all models implies score_new_i - score_base_i = c-c = 0 for every sample i. Any bootstrap resample of zeros has mean 0, so both one-sided bounds are 0 and 0 >= -|tau| holds; by NT20 the gate accepts. (b) If writeback were allowed in this case, the loop would commit arbitrary weights while claiming 'no regression proved', which is false because the proof is vacuous. Rejecting is the unique action that prevents a false claim. (c) Necessity: (a) shows a constant verifier gives a trivial gate. Sufficiency: if V is non-constant, there exist models with a negative paired mean; with enough samples the CI lower bound falls below -|tau| and the gate rejects

(NT15(a)), so the gate is non-trivial. The upstream hooks.py default_verifier returns exactly 0.0 and loop.py:335 checks verifier_is_default before writing. QED

Boundary. Sufficiency in (c) is conditional on bootstrap consistency and sufficient sample size (NT15(a) premises). The theorem says nothing about whether any particular real verifier is well specified.

Architectural use. Formalizes a source-level guard that was previously only a comment, and closes the 'RSI gate is decorative' objection with a proof. *Source:* dream/hooks.py default_verifier; dream/loop.py VERIFY segment

A.22 NT22 - Irreducible gap of hidden-state alignment objectives (TS2SC)

Premises. Teacher is a content-addressed attention block; student is a stack of L static-LTI SGC mixers with a LINEAR residual stream (per-position FFN does not mix history tokens with the query before the final read-out); the alignment objective minimizes the residual-stream cosine to the teacher layer; content-addressing task as in NT13.

Claim. There exists epsilon_0 > 0, INDEPENDENT of L and of the student parameters, such that the alignment objective cannot be driven to zero: on some inputs the achievable residual cosine is at most 1 - epsilon_0. Adding layers does NOT close the gap, because each mixer preserves the premise that history enters only through a fixed linear summary. Hence 'fit the attention internals' is not a tuning problem but a function-class obstruction.

By NT13, a single static-LTI mixer with a linear residual stream cannot implement content addressing: there exist two value assignments v, v+Delta-v giving the same history summary A and the same query q, hence the same student output for any read-out, while the teacher's attention output differs by at least delta = ||E(v_{j1})-E(v_{j2})|| > 0. Therefore the student-teacher output discrepancy is at least delta on one of the two inputs for every parameter setting; this gives epsilon_0 > 0 after normalizing by the bounded residual norms. Stacking: the residual stream of L layers is a sum of layer outputs, each a static-LTI function of the same history plus per-position terms; under the linear-stream premise their SUM is again a fixed linear summary of the history (linearity of the sum), so NT13 still applies verbatim to the final stream. Consequently epsilon_0 does not depend on L. The measured ceiling 0.2484 vs oracle 0.2476 is a finite-sample ESTIMATE of this gap, not a theorem. QED

Boundary. Requires a LINEAR residual stream; per-position nonlinearities that mix A and q before the final read-out leave the premise (this is exactly conjecture C3, unproved). Does not forbid OUTPUT-distribution distillation (NT23), nor does it assert any specific numeric cos.

Architectural use. The formal reason the TS2SC weight-transfer study could not succeed and why the 0.248 ceiling was 'eaten full', not 'one more coef tuning away'. It converts a project-level failure into a theorem. *Source:* TS2SC transfer study + NT13 + O1

A.23 NT23 - Separation of output-distribution distillation from hidden-state alignment

Premises. CONDITIONAL. (H1) The student's logit function class F_S is dense, in KL divergence, in the teacher's output distribution p_T on its support; (H2) the hidden-state alignment objective is as in NT22; teacher is frozen, distillation uses per-position KL with T>=1.

Claim. Under H1, inf over student parameters of KL(p_T || p_S) = 0, while the hidden-state alignment objective has the strictly positive irreducible gap epsilon_0 of NT22. Hence the two objectives are NOT equivalent: output distillation is not blocked by NT13/NT22, whereas internal alignment is. This holds WITHOUT assuming the student's internal states resemble the teacher's.

H1 states that for every eta>0 there exists f in F_S with KL(p_T||softmax(f/T)) < eta; taking eta->0 gives infimum 0. By NT22 the internal-alignment gap is >= epsilon_0 > 0 for all parameters, so its infimum is >= epsilon_0 > 0 = inf KL. Since the two infima differ, the objectives cannot be equivalent: no parameter sequence drives both to zero, and optimizing internal alignment cannot be justified as a surrogate for output KL. The proof uses only H1 and NT22; no claim about the student's hidden states is made. QED

Boundary. CONDITIONAL on H1 (density), which is NOT proved here and may fail for a narrow student logit class; the theorem is a SEPARATION, not a guarantee that KL reaches 0 in finite training. It does not claim output distillation succeeds empirically (O3 is a 3.07M-token pilot).

Architectural use. Gives the migration protocol a principled target: align the teacher's OUTPUT distribution, never its internals. Explains why the earlier ' + ' was the wrong objective. *Source:* TRANSFER_DISTILL_REPORT.md + NT22

A.24 NT24 - Hard stagnation of zero-order self-evolution (minimal-round b

Premises. NT15 holds: at most w insights of b bits are committed per round, for T rounds, so I(history; weight increments) <= T*w*b bits. A task requires the weights to encode a sufficient statistic carrying I_task bits about the L-token history; the final read-out depends on the weights and the current query only.

Claim. (a) If I_task > T*w*b, the weights CANNOT encode a sufficient statistic for the task, so no read-out attains the task's Bayes error: self-evolution is HARD-STAGNATED before round T. (b) The minimum number of rounds satisfies T_min >= ceil(I_task/(w*b)), INDEPENDENT of L and of the bank capacity C. (c) Longer context does not shorten T_min; it only increases the retrievable history at inference, consistent with NT15(b).

(a) Let H be the history, Theta_T the weights after T rounds, and Y the task answer. If the weights encoded a sufficient statistic for Y then I(H;Theta_T) >= I(H;Y) >= I_task. But NT15(b) gives I(H;Theta_T) <= T*w*b < I_task, contradiction. Hence the Markov chain H -> Theta_T -> Y has bottleneck capacity below I_task and cannot achieve the Bayes-optimal error (data processing). (b) Rearranging T*w*b >= I_task gives T >= I_task/(w*b), and T is an integer. (c) Neither L nor C appears in the bound; by NT14(ii) only k selected items influence a read, so the unselected L-k items do not enter the rate. QED

Boundary. An INFORMATION-THEORETIC necessary bound, not a training-convergence claim and not an assertion that the loop reaches the rate. I_task is a property of the TASK, which must be estimated, not measured by accuracy. Does not forbid few-shot in-context retrieval (that uses the episodic bank, NT14).

Architectural use. Answers 'can long-context self-evolution keep improving?' with a provable NO beyond T_min, and converts RSI enthusiasm into a checkable budget inequality. *Source:* CT7(b) / NT15 + NT14(ii)

A.25 NT25 - Tightness of the quantization-resolvable radius

Premises. CONDITIONAL on a uniform-step absolute quantizer with step s, input bound |u_t| <= U, r = s/(2U) < 1, first-order state s_t = rho s_{t-1} + u_t with 0<rho<1; a contribution at delay tau is resolvable iff rho^tau U > s/2.

Claim. (a) The resolvable radius is exactly tau* = ln(r)/|ln rho| (floor), and (b) it is tight: there exists an input with a single unit pulse such that the contribution at tau*+1 is exactly at the quantization threshold and cannot be resolved, while all delays <= tau* are resolvable. (b) tau* = Theta((m+1)/|ln rho|) for m-bit mantissas, so the asymptotic rate is OPTIMAL up to a constant: no deterministic rescaling improves the EXPONENT. (c) For rho = 1-epsilon, tau* ~ (m+1) ln 2 / epsilon, giving the observed order of magnitude of the bf16/int8 long-length drop.

(a) rho^tau U > s/2 iff tau ln rho > ln(s/(2U)) = ln r, and since ln rho<0 this is tau < ln r / ln rho = ln(r)/|ln rho|. The largest integer below it is tau*. Choose the input with u_0=U and all other u_t=0: the state at delay tau is exactly rho^tau U, so at tau*+1 it equals or falls below s/2 by construction, hence is not resolvable; for every tau<=tau* it strictly exceeds s/2. No scaling of U or s changes r's logarithm by more than an additive constant.

(b) With s/U ~ 2^{-(m+1)}, ln(1/r) = (m+1) ln 2 + O(1), so tau* = ((m+1) ln 2 + O(1))/|ln rho|. The dependence on |ln rho| in the DENOMINATOR is forced by the exponential decay law, so no deterministic rescaling alters the exponent; it can only change the O(1) numerator.

(c) |ln(1-epsilon)| = epsilon + O(epsilon^2), giving tau* ~ (m+1) ln 2 / epsilon. QED

Boundary. CONDITIONAL on a uniform absolute quantizer; the physics

probe uses dynamic per-step scaling, so MEASURED drop points (L=2049/4097) are NOT derived from this theorem -- only the order of magnitude is. Tightness is within this model, not for all quantization schemes.

Architectural use. Upgrades the earlier 'conservative sufficient radius' into a TIGHT radius with an explicit precision-vs-decay rate, and explains NT07's interior optimum quantitatively. *Source:* Appendix A.7 perturbation bound + O6 physics probes + NT07

A.26 NT26 - Causal optimality of per-step peak calibration

Premises. CONDITIONAL. Streaming state s_t, causal information set at step t contains only I_t = { |u_s| <= U, s_{t-1}, rho }, i.e. no future peak is known. A quantizer chooses a scale Delta_t >= |s_t| to avoid overflow and incurs absolute error delta_t proportional to Delta_t. A 'fixed global scale' uses Delta_t = Delta for all t.

Claim. (a) The per-step peak rule Delta_t = max(|s_t|, Delta_min) is the POINTWISE MINIMAL admissible scale given I_t, hence minimizes delta_t at EVERY step. (b) Any fixed global scale must satisfy Delta >= sup_{t<=L}|s_t|, and since sup >= max_t|s_t| pointwise, delta_fixed >= max_t delta_peak(t), with strict inequality whenever the peak varies. (c) Therefore fixed-scale quantization is never better and is strictly worse on any sequence with a non-constant state envelope; the improvement is CAUSAL (needs no future information).

(a) Admissibility requires Delta_t >= |s_t|; the smallest such value is |s_t| (or Delta_min for the zero state), which uses only s_{t-1}, u_t and rho -- all in I_t. Minimality is pointwise, so delta_t = c Delta_t is minimized pointwise.

(b) A fixed Delta must cover every t<=L, so Delta >= sup_t |s_t| >= |s_t| = Delta_peak(t) for each t; hence delta_fixed >= delta_peak(t) for all t, i.e. >= max_t delta_peak(t). If the envelope is non-constant the inequality is strict at the step where |s_t| < sup.

(c) Combining (a) and (b) gives the partial order, and (a) shows the rule is implementable causally. QED

Boundary. CONDITIONAL on scale-proportional absolute error and the stated causal information set. It proves a PARTIAL ORDER of error bounds, NOT that dynamic scaling always wins empirically (O6's fp16 non-monotonicity is a separate genuine numerical effect). It does not contradict the withdrawal of 'dynamic scaling causes spurious artifacts' -- those are different claims.

Architectural use. Upgrades the previously DEMOTED one-line partial order (T14) into a causal-optimality theorem, and is exactly the standard that separates 'too ordinary to write' from 'non-trivial enough'. *Source:* O6 fixed-scale controls + T14 + docs physics frontier

A.27 NT27 - Phase transition of the optimal decay across precision depths

Premises. CONDITIONAL. NT07's resolvable-length bound g(rho) = ln((1-rho)U/(c delta)) / |ln rho| on rho in (0, hi), hi = 1 - c delta/U; delta/U = 2^{-(m+1)} for mantissa depth m; c>1 fixed; delta/U sufficiently small.

Claim. (a) g is strictly increasing near rho -> 0 and strictly decreasing near hi, so for small delta/U it has a SINGLE interior maximum rho*(m). (b) rho*(m) solves the stationarity equation (1-rho) ln(1/rho) ln((1-rho)U/(c delta)) = rho ln(1/rho) + ... equivalently the balance (1-rho) ln((1-rho)U/(c delta)) = rho ln(1/rho); asymptotically rho*(m) = 1 - Theta(2^{-m}). (c) Therefore DIFFERENT precision depths have DIFFERENT optimal decays: a single rho tuned for fp32 is provably suboptimal for int8. rho -> 1 is never optimal (NT07).

Write N(rho)=ln((1-rho)U/(c delta)) > 0 and D(rho)=ln(1/rho)>0, g=N/D. Then g' = (N' D - N D')/D^2 with N' = -1/(1-rho) < 0 and D' = -1/rho < 0, so sign(g') = sign(-D/(1-rho) + N/rho). As rho -> 0, D = ln(1/rho) -> +infinity but N -> ln(U/(c delta)) is finite and positive; the term N/rho ~ ln(U/(c delta))/rho grows as 1/rho and therefore dominates -D/(1-rho) ~ -ln(1/rho). Hence g' > 0 near 0. As rho -> hi, N -> 0+ while D stays positive and finite; then N/rho -> 0 and -D/(1-rho) < 0 dominates, so g' < 0 near hi. By continuity g' has a zero in (0,hi); for small delta/U the ratio N/D is unimodal (one sign change), giving a UNIQUE interior maximum rho*. Setting g'=0 gives N/rho = D/(1-rho), i.e. (1-rho) N = rho D, the stated balance. Now put rho = 1-epsilon with epsilon small and delta/U = 2^{-(m+1)}: then D = ln(1/(1-epsilon)) = epsilon + O(epsilon^2) and N = ln(epsilon * 2^{m+1}/c) = (m+1) ln 2 + ln epsilon - ln c. The balance (1-rho)N = rho D becomes epsilon * N = (1-epsilon) * (epsilon + O(epsilon^2)), so for epsilon>0 we need N = 1 + O(epsilon). Substituting N gives ln epsilon = 1 - (m+1) ln 2 + ln c + O(epsilon), hence epsilon = c*e*2^{-(m+1)} (1+o(1)) = Theta(2^{-m}) and rho* = 1 - Theta(2^{-m}). This matches the NT07 finite scans (m=8 -> 1-rho* ~ 1e-2, m=11 -> ~1e-3, m=24 -> ~3e-8). Since rho* depends on m, a single rho cannot be optimal for two different mantissa depths. QED

Boundary. CONDITIONAL on NT07's uniform-error model; uniqueness of the stationary point is asserted for SMALL delta/U (the regime of interest); it is not a global proof for all parameters. The asymptotic rho*=1-Theta(2^{-m}) is analytic, and is CROSS-CHECKED against NT07's finite scans but not derived from them. Measured O6 rho=0.999 is an observation.

Architectural use. Explains why the int8 / bf16 / fp32 frontiers must be tuned separately and gives the precision-dependent decay law; strengthens NT07 from 'an interior optimum exists' to 'how it moves with m'. *Source:* NT07 + O6 physics_precision.json

A.28 NT28 - Layer x tensor-order incompatibility inequality for H-hop retr

Premises. A stack of L layers; each layer's output is a polynomial of total degree at most the tensor order o of its state contraction (order-2 outer product gives o=2; bilinear contraction degree 2). Skip connections do not raise degree. The H-hop target contains H key-match factors plus one value factor, total degree H+1.

Claim. L * ln(o) >= ln(H+1), equivalently o^L >= H+1. For o=2 this recovers NT11 (L >= log2(H+1)); for L=1 it forces o >= H+1, so a SINGLE layer needs tensor order at least H+1. CRITICALLY: increasing the number of pole channels P changes the KERNEL SHAPE but not the tensor order o, so P cannot compensate for too few layers or too low an order. The two resources (depth, order) are incomparable and must both satisfy the product bound.

Composition of polynomials of degrees d_1..d_L has degree at most prod d_i <= o^L; skip connections (addition) do not increase degree. The H-hop target is a multilinear form of degree H+1 (H factors <phi(q),psi(k)> plus one value). A polynomial of degree < H+1 cannot equal a nonzero degree-(H+1) form, so o^L >= H+1, giving the inequality. Setting o=2 gives NT11; setting L=1 gives o>=H+1. The pole count P enters only through K_n(d)=sum_p w_{p,n} rho_{p,n}^d, a SCALAR multiplier of each delay; it does not add a new tensor index or raise the polynomial degree in the history variables. Hence P is orthogonal to the bound. QED

Boundary. NECESSARY condition only; no sufficient construction. The degree bound is an upper bound on what L layers of order o can reach. Per-position nonlinearities of degree > o change the premise. The measured H4_multihop=0.0322 is a finite-sample observation.

Architectural use. Unifies NT01 (order 2 for 1 hop) and NT11 (layers for H hops) into one inequality, and PROVES that the 'add more poles' fix cannot work -- closing a whole search direction that was tried implicitly. *Source:* NT01 + NT11 + O5 H4_multihop

A.29 NT29 - Topological necessity of a discontinuous hard pointer

Premises. Exact retrieval map R(q) = v_{j*(q)} with j*(q) = argmax_j <phi(q), k_j>, where >=2 distinct keys k_1,k_2 have distinct values v_1 != v_2 and the key space is a continuum containing the bisector hyperplane between them. The state is constant-size (no per-token cache).

Claim. (a) R is DISCONTINUOUS on the bisector, hence not differentiable there; hard pointer selection is therefore a NECESSARY feature of exact retrieval, not an implementation artifact. (b) Any CONTINUOUS surrogate Rtilde that agrees with R off an epsilon-neighborhood of the bisector has a NONZERO mis-selection probability bounded below by the measure of that neighborhood under any query distribution with positive density there. (c) Softening cannot simultaneously be continuous AND exact; this is an INDEPENDENT obstruction from NT14(iii)'s kernel-collision argument (which concerns the aggregate read-out, not the selector).

(a) Let q_1,q_2 be points on opposite sides of the bisector B = {q : <phi(q),k_1> = <phi(q),k_2>} with <phi(q_1),k_1> > <phi(q_1),k_2> and

the reverse at q_2. Then R(q_1)=v_1 and R(q_2)=v_2. Take a path q(s) from q_1 to q_2 crossing B at s0; the left limit of R is v_1 and the right limit is v_2 != v_1, so R is discontinuous at s0 and has no derivative there.

(b) A continuous Rtilde must pass from v_1 to v_2 continuously, so on the epsilon-neighborhood N of B it takes values strictly between them; no point of N is mapped to the exact correct value v_1 or v_2. Hence P[Rtilde(q) != R(q)] >= P[q in N] > 0 for any query law with positive density on N.

(c) NT14(iii) shows a differentiable SOFT AGGREGATE over all slots has a kernel collision; (a)-(b) show the SELECTOR itself cannot be continuous. They are different maps with different proofs, so the two obstructions are independent. QED

Boundary. Requires a continuum query space and >=2 distinguishable keys; a finite/discrete query alphabet could in principle use a lookup table (but that is itself a discontinuous map on the embedding space). Does not forbid approximate retrieval (NT08).

Architectural use. Gives the L0/L1/L2 three-tier sidecar its theoretical justification: the L2 top-k POINTER must be hard/non-differentiable, which is why it sits outside the differentiable SGC mixer and cannot be folded into end-to-end training. *Source:* docs L0/L1/L2 tier design + memory.py topk + NT14

A.30 NT30 - End-to-end tightness chain: none of the five bounds can be improved

Premises. Exact H-hop retrieval over V token classes, N value dimensions, M second-order basis functions, in a static-LTI constant-state architecture with no N x N scoring and no growing KV cache; NT01, NT04, NT05, NT19, NT28 and NT13 all hold.

Claim. Any such architecture simultaneously requires (i) tensor order o >= H+1 for L=1, or o^L >= H+1 in general (NT28); (ii) code dimension R >= V (NT04); (iii) state >= 2*M*R*N scalars, with R < 2V attainable (NT05, NT19); (iv) a strictly positive kernel, hence real-pole roots (NT16/NT01). TSC at H=1 attains ALL FOUR within constant factors: order 2, R<2V, 2M*R*N<4M*V*N, real poles. Moreover the chain is JOINTLY TIGHT: improving any one bound alone cannot lower the product, because each is forced by a different independent theorem.

Chain the results: NT13 forbids order 1; NT01 achieves order 2 (tight). NT04(a) gives R>=V and NT04(b) gives R<2V (2-optimal, tight within the class). NT19 gives 2M scalars per code coordinate per value channel as a MINIMAL sufficient statistic, so with R code rows and N value channels the state is >= 2MRN; TSC uses exactly P*R*N with P=M poles, i.e. M*R*N first-order registers, and the second-order version 2MRN is attained by the recurrence (tight). NT16 forces strict positivity, hence real poles by NT02. For H hops NT28 forces o^L >= H+1, which no change in P, R, M or N can relax. Each inequality comes from a DIFFERENT proof (rank-nullity / orthogonality / Hankel rank / Turan / degree composition), so relaxing one would contradict its own theorem without affecting the others: the product bound is jointly tight. TSC at H=1 meets all four simultaneously, so the bounds are not merely lower bounds but ACHIEVED ones. QED

Boundary. Tightness is up to constant factors (the factor 2 in R<2V, and the pole count P=M convention). It does NOT assert that no architecturally DIFFERENT construction (e.g. a non-LTI state) could beat the product; that would leave the premise. Neither LM parity nor throughput is implied.

Architectural use. The paper's single strongest summary statement: instead of one accuracy number, an end-to-end achieved tight bound. It also PROVES the project cannot be improved by 'add more poles / bigger grid / better code' alone -- only by leaving the premise (NT28). *Source:* NT01/NT04/NT05/NT16/NT19/NT28 consolidated

Appendix B Conjectures and Open Questions

Explicitly unverified. None is a premise of any theorem or a claimed result. NT04 closes C2; NT11 gives C1/C3 a necessary condition but no construction.

· **C1.** Stacking TSC layers or inter-layer recurrence restores multi-hop retrieval. NT11 gives the layer lower bound; no stacking experiment exists.

· **C2.** CLOSED by NT04: a deterministic orthogonal code exists for every V with R < 2V.

· **C3.** NT13 extends to deep stacks with per-position nonlinearities. Cross-terms between A and q leave the fixed-summary premise; unproved.

· **C4.** Freezing rho = 0.999 from the start of training remains stable on >=10M-token language modeling. Verified only on a synthetic task.

· **C5.** Layered quantization or residual accumulation pushes the int8 long-length boundary further. NT07 bounds the trade-off; untested schemes remain.

· **C6.** On equal compute the periodic and content-addressing axes are complementary. The periodic-axis gain was NOT replicated (O4).

· **C7.** CLOSED IN PART by NT03: the rho<1 multi-pole kernel is provably not the linear-attention constant kernel. Full-model separation from all gated variants remains open.

· **C8.** The decay spectrum gives an inductive bias on oscillatory/damped physical signals that attention lacks. No measured comparison exists.

Appendix C Falsified Statements

· **F1.** Increasing rho_max extends the effective memory window. Rejected by measurement.

· **F2.** State-feedback rho_t/theta_t modulation (SFSC) preserves FFT duality. Rejected: LTV is non-Toeplitz (NT12); FFT-proxy relative error 0.9720.

· **F3.** Phase-resonance slots exceed equal-parameter attention. Rejected (O2).

· **F4.** Content addressing and FFT/IIR duality are mutually exclusive. WITHDRAWN via NT01/NT10; this withdrawal is a core contribution.

· **F5.** An oscillatory basis itself provides the inductive advantage. Rejected: a random orthogonal basis reached 1.0000; NT02/NT08 explain why.

· **F6.** Dynamic quantization scaling causes spurious non-monotonicity. Rejected by same-sample prefix controls.

· **F7.** Shannon capacity limits long-length retrieval here. Rejected: 176x headroom; NT07 identifies the decay/precision trade-off instead.

· **F8.** A monotone verifier score implies monotone self-improvement (former T16). WITHDRAWN (W7); trivial scalar induction and inconsistent with the multi-metric paired-bootstrap gate. Replaced by NT15.

Appendix D Observation Provenance

All values were computed from raw result files accompanying the anonymized supplementary artifact; none was transcribed from a secondary summary.

Obs.	Config / seeds	Key values	File
O1	1.7B transfer, 1 run	0.2484 / 0.2476 / 0.5430	transfer study
O2	single-hop, 3 seeds	0.8672 vs 1.0000	phase-resonance probe
O3	470K, 3.07M tokens	10.817 / 11.281 / 10.217	native pilot
O4	7 grids, 1 run	0.1094-0.1354 vs 0.1575	periodic-axis probe
O5	V=64, 5 seeds, 512/seed	1.0000 / 0.9987 / p=0.4998	TSC probe
O6	5 seeds, 4 precisions	L=16385, 43/43 checks	physics probes

Measured configuration details. O5: W1 has 10264 parameters, W4_half 4920, M5 2792; the stored reference record has 10240. **Fisher p = 0.4998 means no significant difference was detected, NOT equivalence.** The paired bootstrap on the general second-order dictionary gave mean difference 0.1901 with 95% CI [0.1582, 0.2220] over 3 seeds (sample-internal variance only). O6: frozen-rho retraining held-out means [1.0, 1.0, 1.0, 1.0], between-seed SDs [0.0, 0.0, 0.0, 0.0]; L = 16,385 accuracy mean 1.0000, min 1.0000; per-token cost at L = 4097 was 35.25 us with a 56320 B state. Conservation guards reported FFT/IIR relative errors of order 1e-16.

Appendix E Reproducibility and Tool Use

Every theorem is derived analytically in Appendix A and requires no execution. Every observation comes from a deterministic script with recorded seeds, precision and sample counts. The supplementary artifact contains raw result files, gradient finite-difference checks, conservation assertions, the 49-item audit (49/49 PASS) and the theorem ledger. Gradient checks passed 8/8, 10/10, 28/28 and 44/44; the physical consistency checker passed 43/43. No training run is used to support any theorem.

Generative AI tools were used for implementation assistance, experiment scripting and language editing. They did not originate the research questions, conjectures or conclusions; no experimental data, citation or result was generated by such a tool. The human author verified every numerical claim against raw files and accepts full responsibility.

Appendix F Withdrawn Claims Register

Earlier working-version statements, formally withdrawn so the revision is auditable. Withdrawals are contributions, not hidden errata.

Withdrawn statement	Reason
Unconditional fp32 bitwise duality	Different rounding orders; real-field only (NT09)
Content addressing <-> duality exclusive	WITHDRAWN, core contribution: NT01/NT10 counterexample
Periodic-axis superiority over attention	Not replicated in-repository (O4)
Equivalent 4-7B parameter claim	No supporting measurement
NT13 holds for any read-out unrestricted	Exactly four premises required; stated in abstract
Dynamic scaling causes spurious effects	Same-sample prefix controls (O6)
W7: monotone score => self-improvement (T16)	Trivial induction + mismatched source gate; replaced by NT15
F8: scalar monotone score as gate	NT20 proves the gate is the unique multi-metric AND rule

Appendix G The 49-Item Audit Ledger

Every statement in this project is adjudicated exactly once. The 30 theorem items have complete analytic proofs in Appendix A; their finite-identity checks only certify correct formation and non-empty premises. The 19 non-theorem items are CLASSIFIED with a reproducible witness or an explicit falsification / experiment specification; PASS for those means 'correctly classified', NOT 'true'. No measured value is a theorem.

Class	Count	Why it is NOT a theorem (if applicable)
New theorems NT01-30	30	Universal, analytic, non-trivial; 6 groups
Falsified F1-F8	8	Explicit counterexamples / witness values
Withdrawn W1-W5, W7	5	Formally retracted earlier working claims
Conjectures C1, C3, C7	3	Unproved; falsification experiment specified
Empirical E1-E3	3	LM parity / SOTA / throughput: no theorem can establish

G.1 The 30 theorems (six de-duplicated groups)

· **I: TSC construction and optimality.** NT01, NT02, NT03, NT04, NT05

· **II: grid, precision and length.** NT06, NT07, NT08, NT09, NT10

· **III: multi-hop, self-evolution, equivalence.** NT11, NT12, NT13, NT14, NT15

· **IV: algebraic structure and identifiability.** NT16, NT17, NT18, NT19

· **V: self-evolution and transfer objectives.** NT20, NT21, NT22, NT23, NT24

· **VI: numerical frontier and multi-hop duality.** NT25, NT26, NT27, NT28, NT29, NT30

G.2 The 19 non-theorem statements

· **F1 [FALSIFIED].** Increasing rho_max extends the window; NT07/NT27 show the optimum is interior and rho->1 shortens resolvable length.

· **F2 [FALSIFIED].** SFSC time-varying rho_t/theta_t preserves FFT duality; time-varying a_t gives non-Toeplitz h (NT12), FFT-proxy error 0.9720.

· **F3 [FALSIFIED].** Phase-resonance slots exceed equal-parameter attention; witness 0.8672 < 1.0000 (5,217 vs 5,152 params).

· **F4 [FALSIFIED].** Content addressing <-> FFT duality mutually exclusive; withdrawn via NT01/NT10. This withdrawal IS a core contribution.

· **F5 [FALSIFIED].** The oscillatory basis itself gives the advantage; random orthogonal code also reaches 1.0000 (NT08), and NT16 shows conjugate poles are not strictly positive.

· **F6 [FALSIFIED].** Dynamic scaling causes spurious artifacts; same-sample prefix controls, and NT26 proves per-step peak is causally pointwise-minimal.

· **F7 [FALSIFIED].** Shannon capacity limits long-length retrieval; 1,280 required bits vs 225,280-bit state (176x headroom), noise 0.20 leaves accuracy unchanged.

· **F8 [FALSIFIED].** A scalar monotone verifier score is the gate; NT20 proves the gate is the UNIQUE multi-metric AND rule.

· **W1 [WITHDRAWN].** Unconditional fp32 bitwise duality; real-field only (NT09).

· **W2 [WITHDRAWN].** Periodic-axis superiority over attention; not replicated (O4).

· **W3 [WITHDRAWN].** Equivalent 4-7B parameter claim; no supporting measurement.

· **W4 [WITHDRAWN].** NT13 holds for any read-out unrestricted; exactly four premises required.

· **W5 [WITHDRAWN].** Dynamic scaling causes spurious effects; same-sample prefix controls (O6).

· **W7 [WITHDRAWN].** Monotone score => self-improvement (T16); trivial induction and mismatched with the source gate; replaced by NT15.

· **C1 [CONJECTURE].** Stacking TSC layers restores multi-hop retrieval. NT11/NT28 give only NECESSARY lower bounds; falsify with L=2,3 stacks on H=2,3, >=5 seeds.

· **C3 [CONJECTURE].** NT13 extends to deep stacks with per-position nonlinearities; cross-terms leave the fixed-summary premise; falsify with a 2-layer FFN-mixed counterexample.

· **C7 [CONJECTURE].** TSC is outside EVERY gated linear-attention variant; NT03 settles only rho<1 multi-pole vs rho=1 single-pole; falsify by exhibiting a gated variant with the same kernel family.

· **E1 [EMPIRICAL].** >=10M-token equal-compute LM parity; requires >=10M tokens, >=5 seeds, paired CI, and same-run Mamba-2/GLA/DeltaNet/Hyena/TS baselines. O3 is a 3.07M-token pilot.

· **E2 [EMPIRICAL].** RSI reaches SOTA / general-language superiority; NT24 gives a necessary rate bound, not capability. No theorem can establish this.

· **E3 [EMPIRICAL].** Absolute throughput advantage over attention; O6 is an unoptimized reference implementation, only scaling is informative; requires a measured same-run comparison.

Why 49 theorems is not the target. Eight statements are demonstrably false, five were formally retracted, three have no proof, and three are empirical claims about scale and capability. Promoting any of them would reintroduce exactly the theorem inflation and 'numerical verification posing as proof' that the withdrawn-claims register (Appendix F) exists to prevent. The competitive contribution is 30 non-trivial theorems plus an auditable account of the 19 statements that must NOT be theorems.